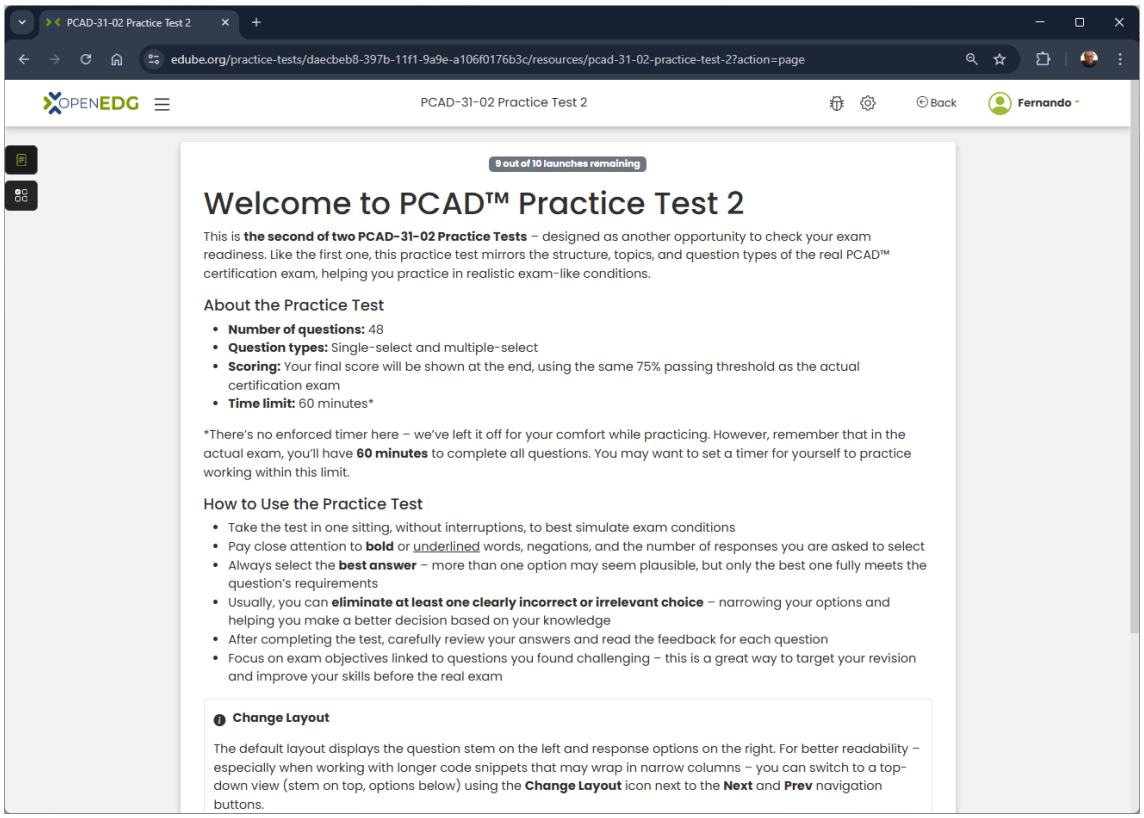
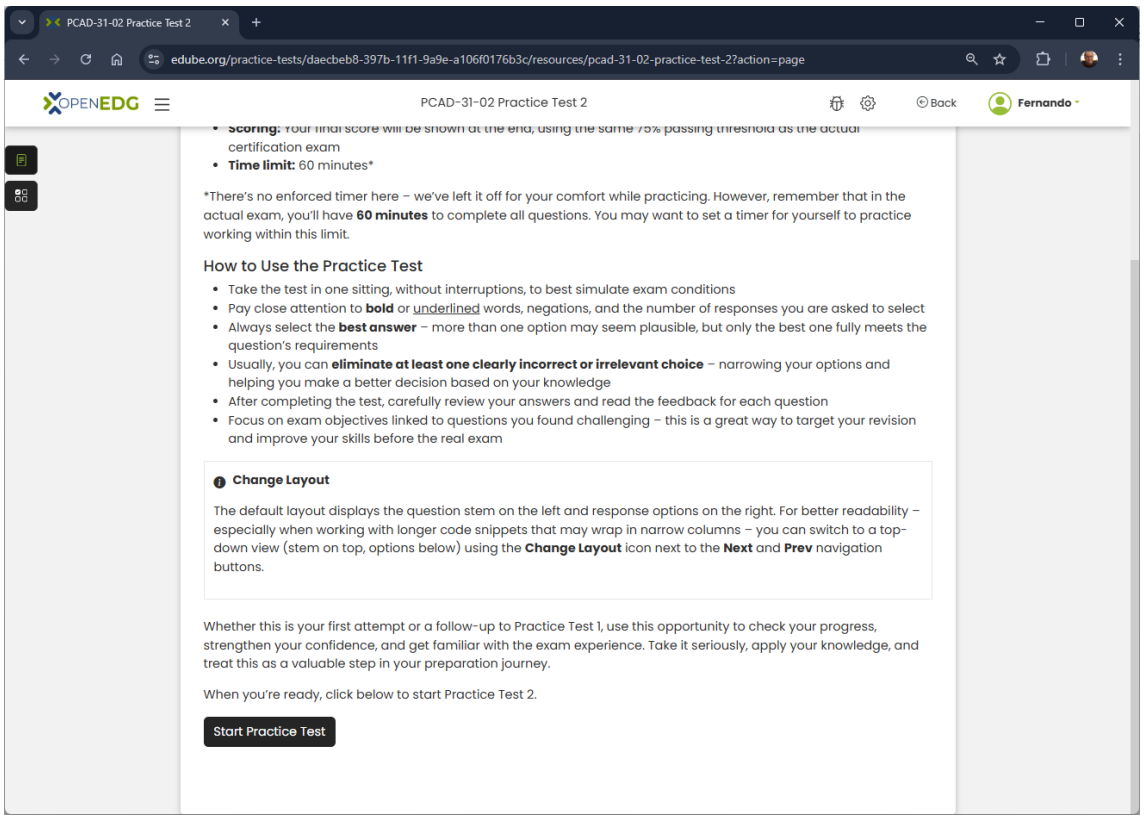
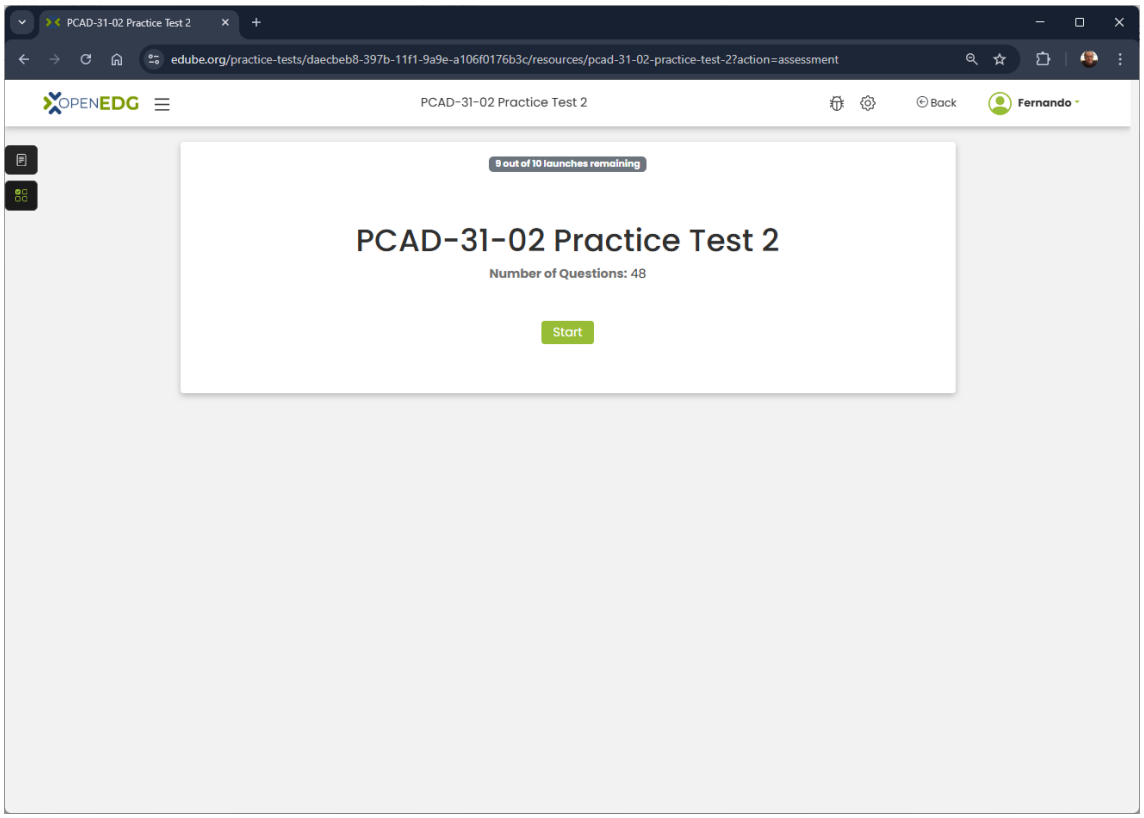
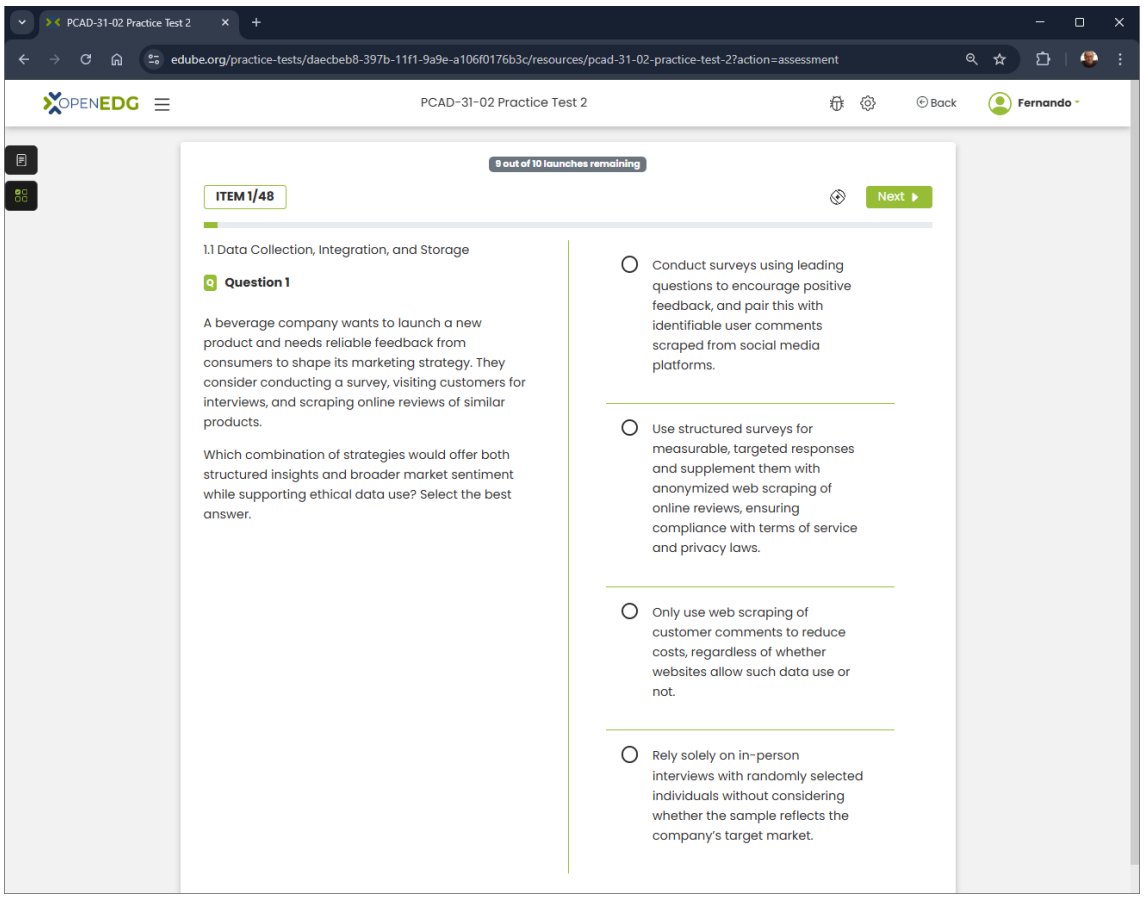
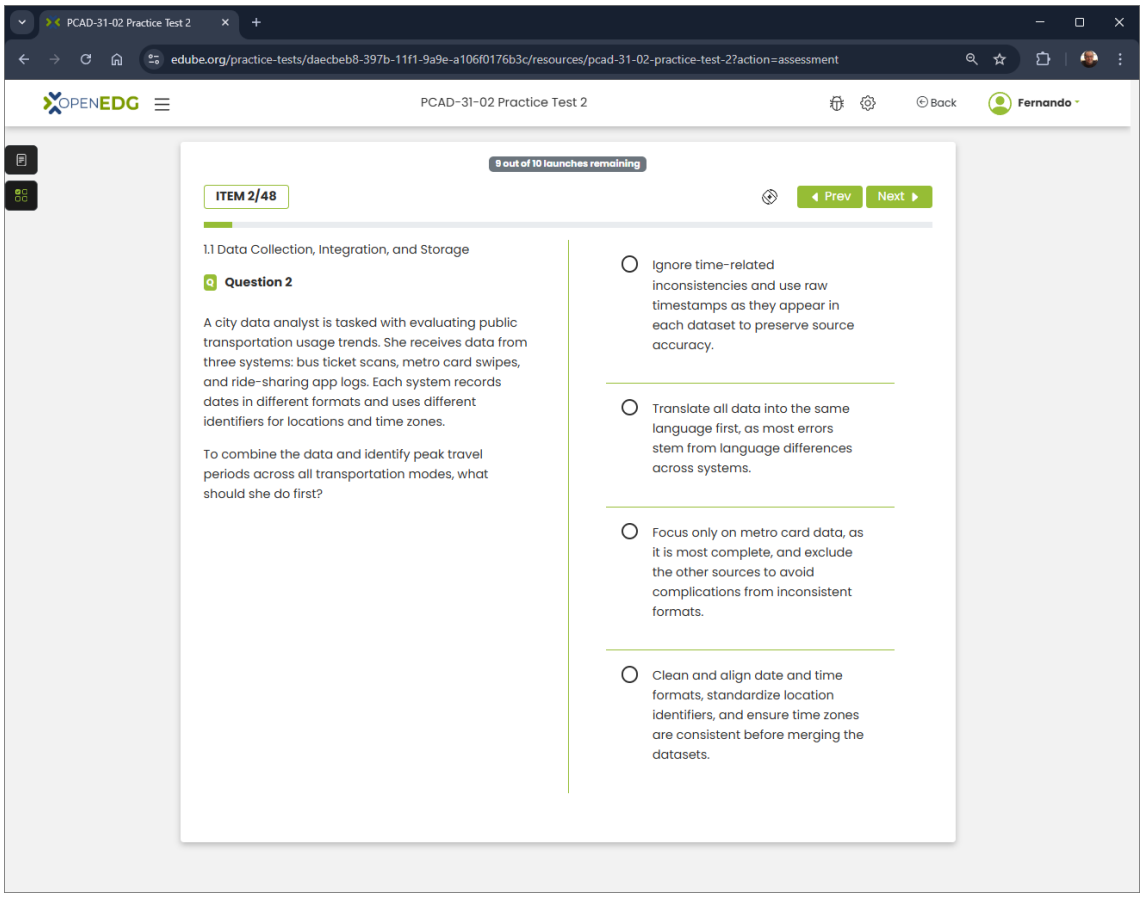
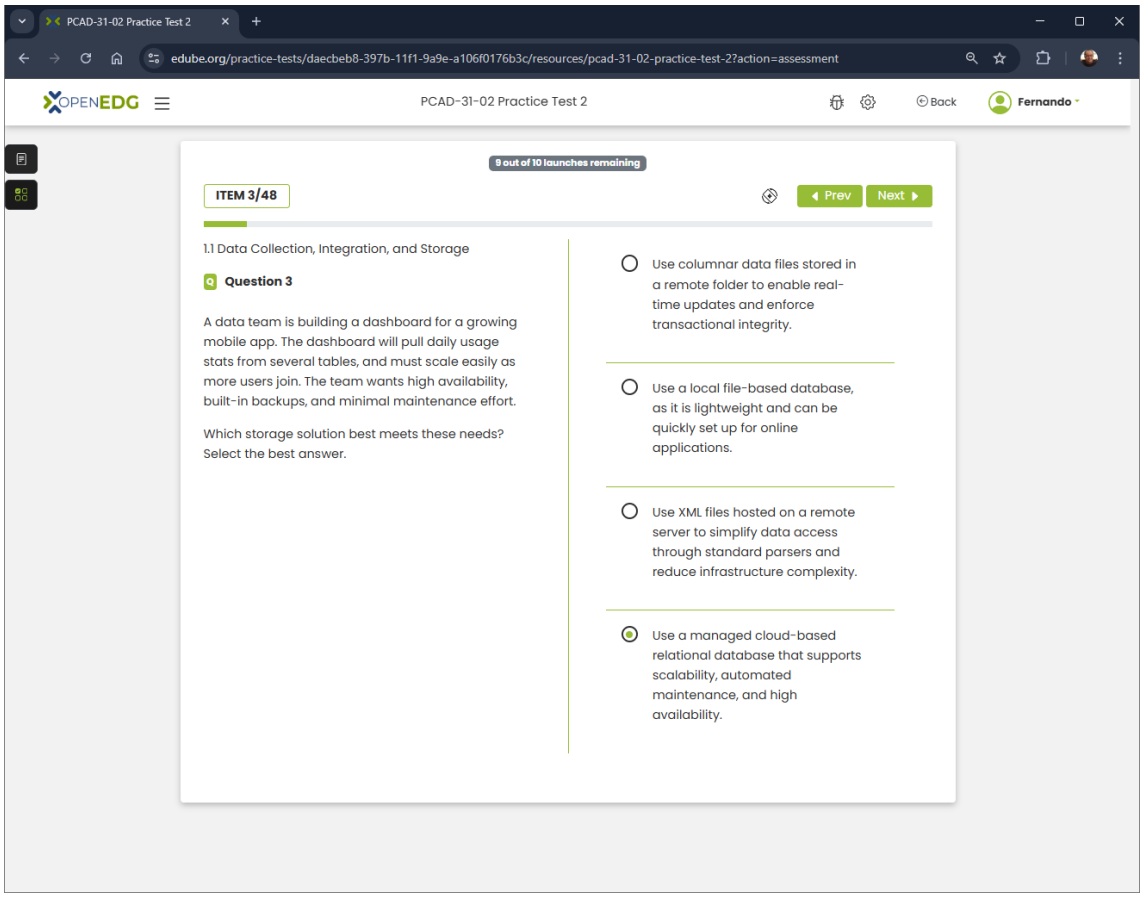

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=page
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
Welcome to PCAD™ Practice Test 2
This is the second of two PCAD-31-02 Practice Tests – designed as another opportunity to check your exam readiness. Like the first one, this practice test mirrors the structure, topics, and question types of the real PCAD™ certification exam, helping you practice in realistic exam-like conditions.
About the Practice Test
Number of questions: 48
Question types: Single-select and multiple-select
Scoring: Your final score will be shown at the end, using the same 75% passing threshold as the actual certification exam
Time limit: 60 minutes*
*There's no enforced timer here – we've left it off for your comfort while practicing. However, remember that in the actual exam, you'll have 60 minutes to complete all questions. You may want to set a timer for yourself to practice working within this limit.
How to Use the Practice Test
Take the test in one sitting, without interruptions, to best simulate exam conditions
Pay close attention to bold or underlined words, negations, and the number of responses you are asked to select
Always select the best answer – more than one option may seem plausible, but only the best one fully meets the question's requirements
Usually, you can eliminate at least one clearly incorrect or irrelevant choice – narrowing your options and helping you make a better decision based on your knowledge
After completing the test, carefully review your answers and read the feedback for each question
Focus on exam objectives linked to questions you found challenging – this is a great way to target your revision and improve your skills before the real exam
Change Layout
The default layout displays the question stem on the left and response options on the right. For better readability – especially when working with longer code snippets that may wrap in narrow columns – you can switch to a top-down view (stem on top, options below) using the Change Layout icon next to the Next and Prev navigation buttons.

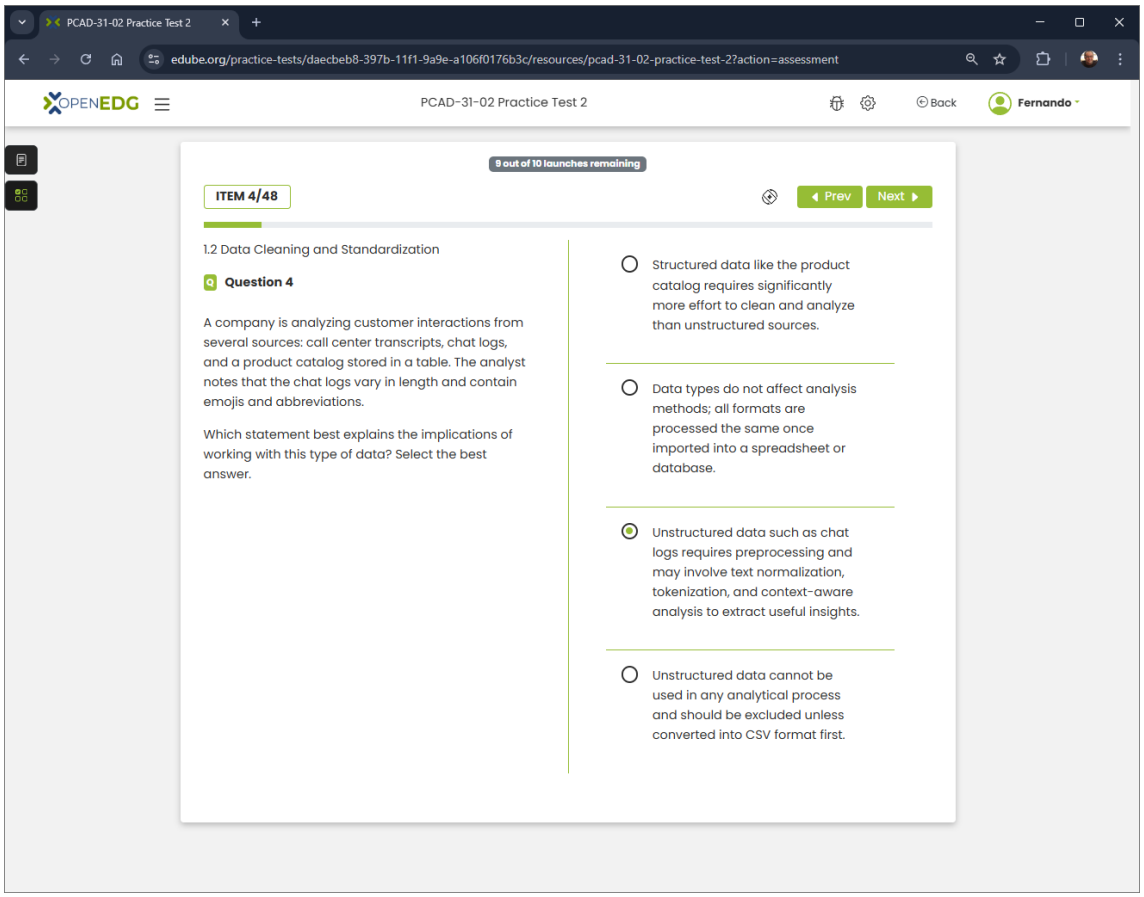

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
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Whether this is your first attempt or a follow-up to Practice Test 1, use this opportunity to check your progress, strengthen your confidence, and get familiar with the exam experience. Take it seriously, apply your knowledge, and treat this as a valuable step in your preparation journey.
When you're ready, click below to start Practice Test 2.
Start Practice Test

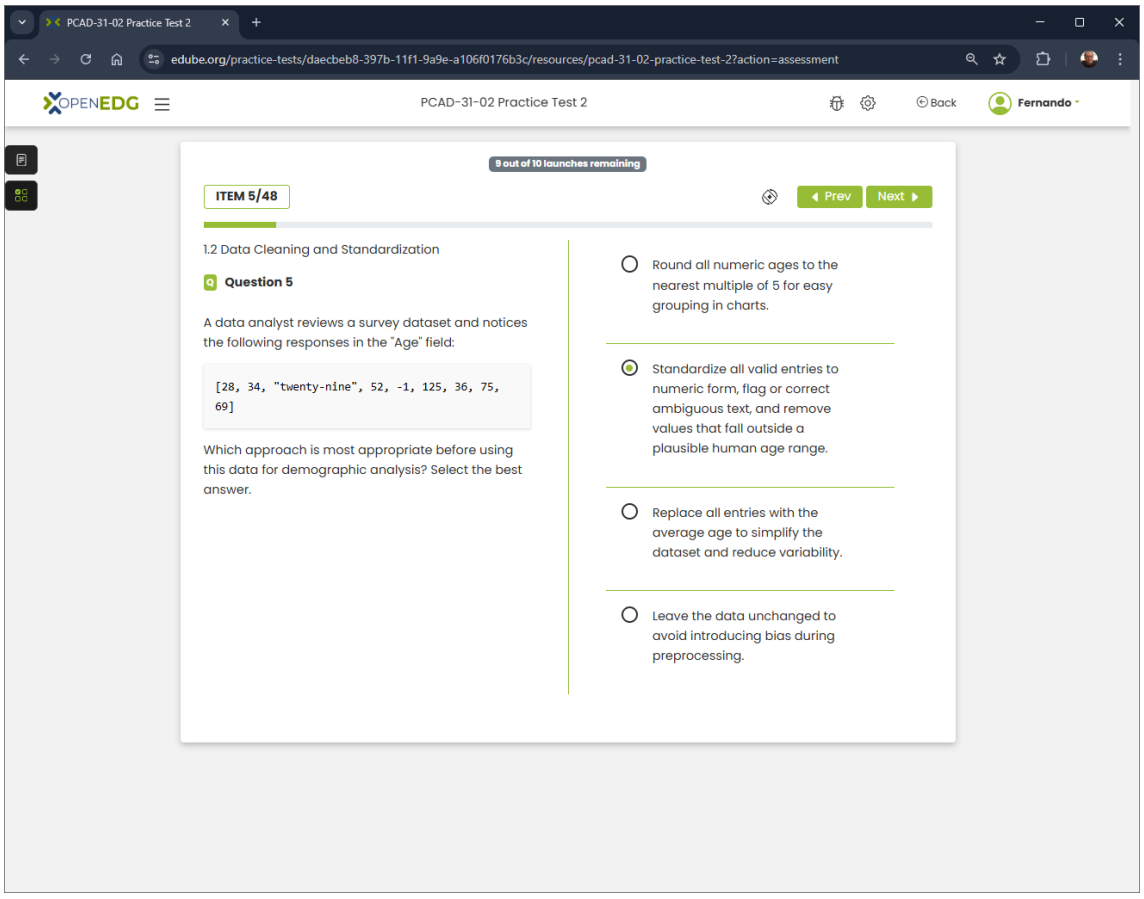

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
PCAD-31-02 Practice Test 2
Number of Questions: 48
Start

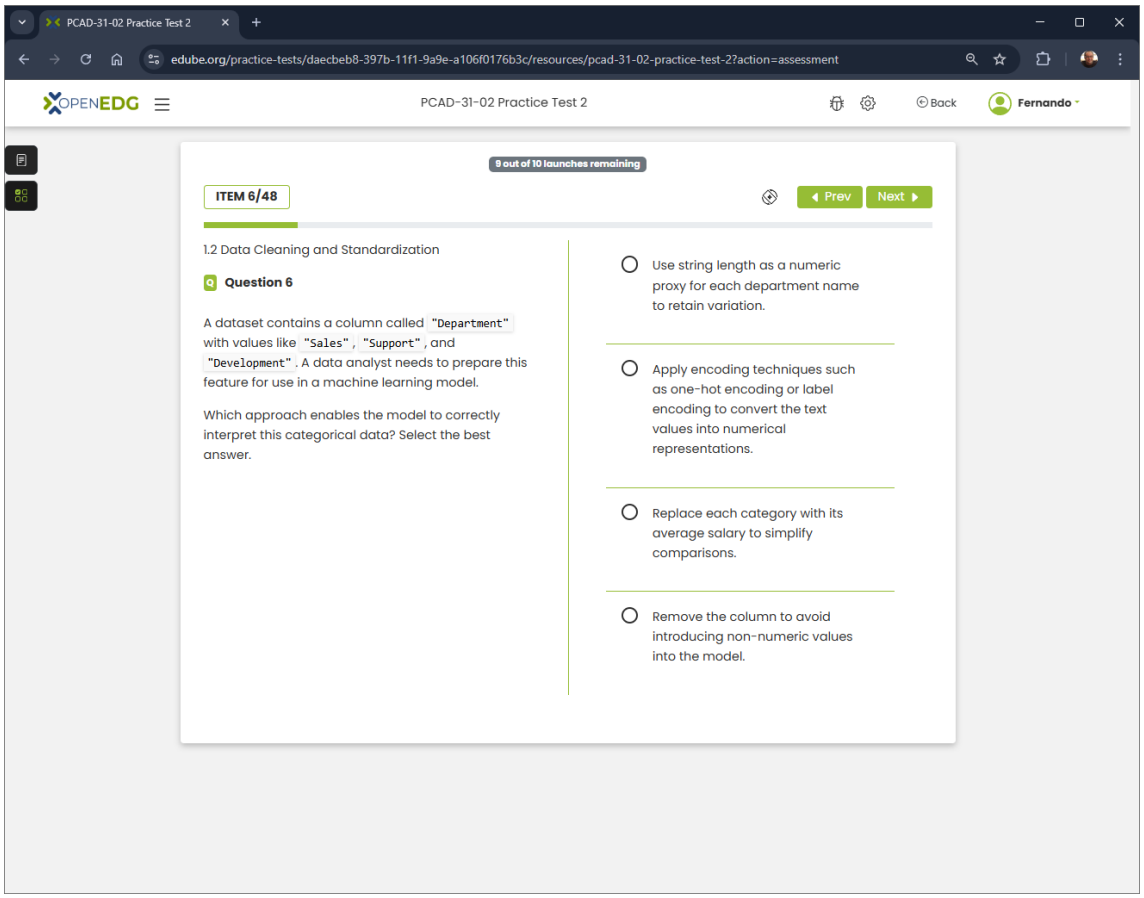

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 1/48
Next
1.1 Data Collection, Integration, and Storage
Question 1
A beverage company wants to launch a new product and needs reliable feedback from consumers to shape its marketing strategy. They consider conducting a survey, visiting customers for interviews, and scraping online reviews of similar products.
Which combination of strategies would offer both structured insights and broader market sentiment while supporting ethical data use? Select the best answer.
Conduct surveys using leading questions to encourage positive feedback, and pair this with identifiable user comments scraped from social media platforms.
Use structured surveys for measurable, targeted responses and supplement them with anonymized web scraping of online reviews, ensuring compliance with terms of service and privacy laws.
Only use web scraping of customer comments to reduce costs, regardless of whether websites allow such data use or not.
Rely solely on in-person interviews with randomly selected individuals without considering whether the sample reflects the company's target market.

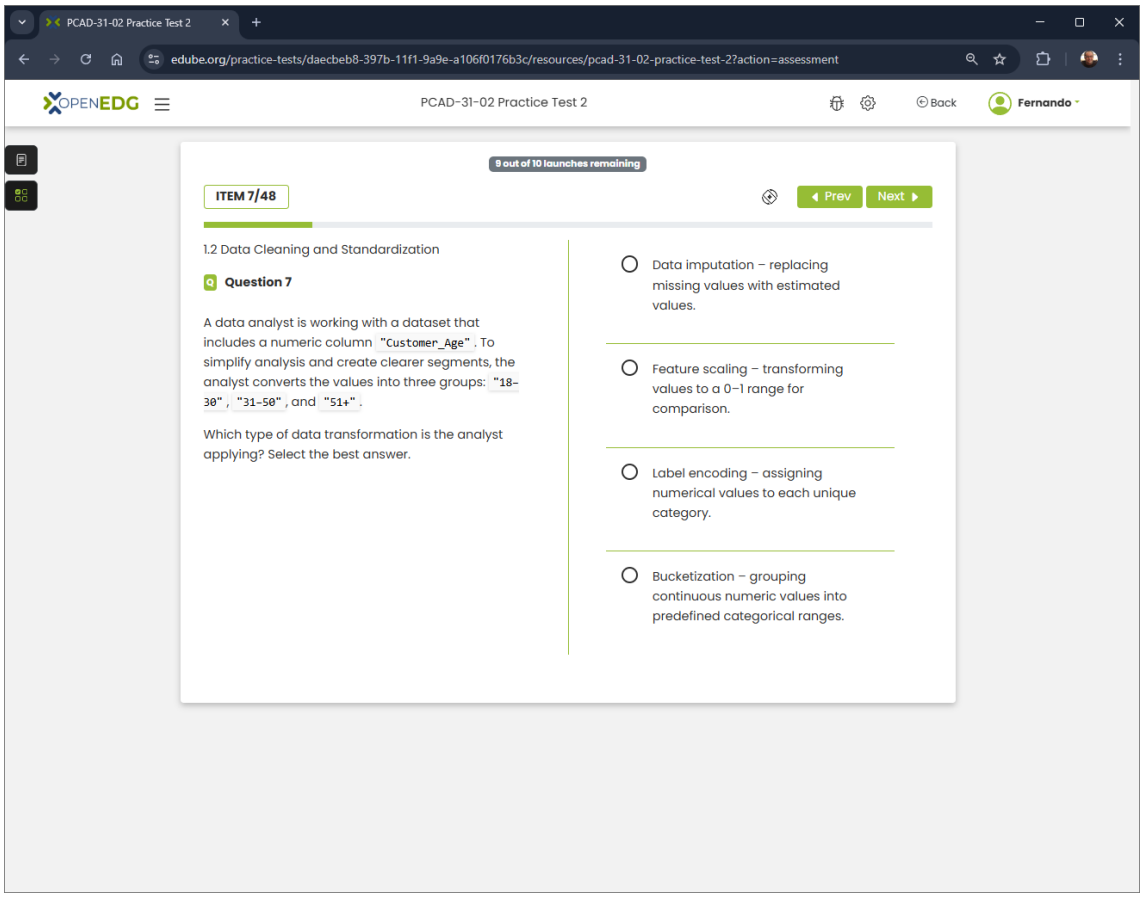

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 2/48
Prev
Next
1.1 Data Collection, Integration, and Storage
Question 2
A city data analyst is tasked with evaluating public transportation usage trends. She receives data from three systems: bus ticket scans, metro card swipes, and ride-sharing app logs. Each system records dates in different formats and uses different identifiers for locations and time zones.
To combine the data and identify peak travel periods across all transportation modes, what should she do first?
Ignore time-related inconsistencies and use raw timestamps as they appear in each dataset to preserve source accuracy.
Translate all data into the same language first, as most errors stem from language differences across systems.
Focus only on metro card data, as it is most complete, and exclude the other sources to avoid complications from inconsistent formats.
Clean and align date and time formats, standardize location identifiers, and ensure time zones are consistent before merging the datasets.

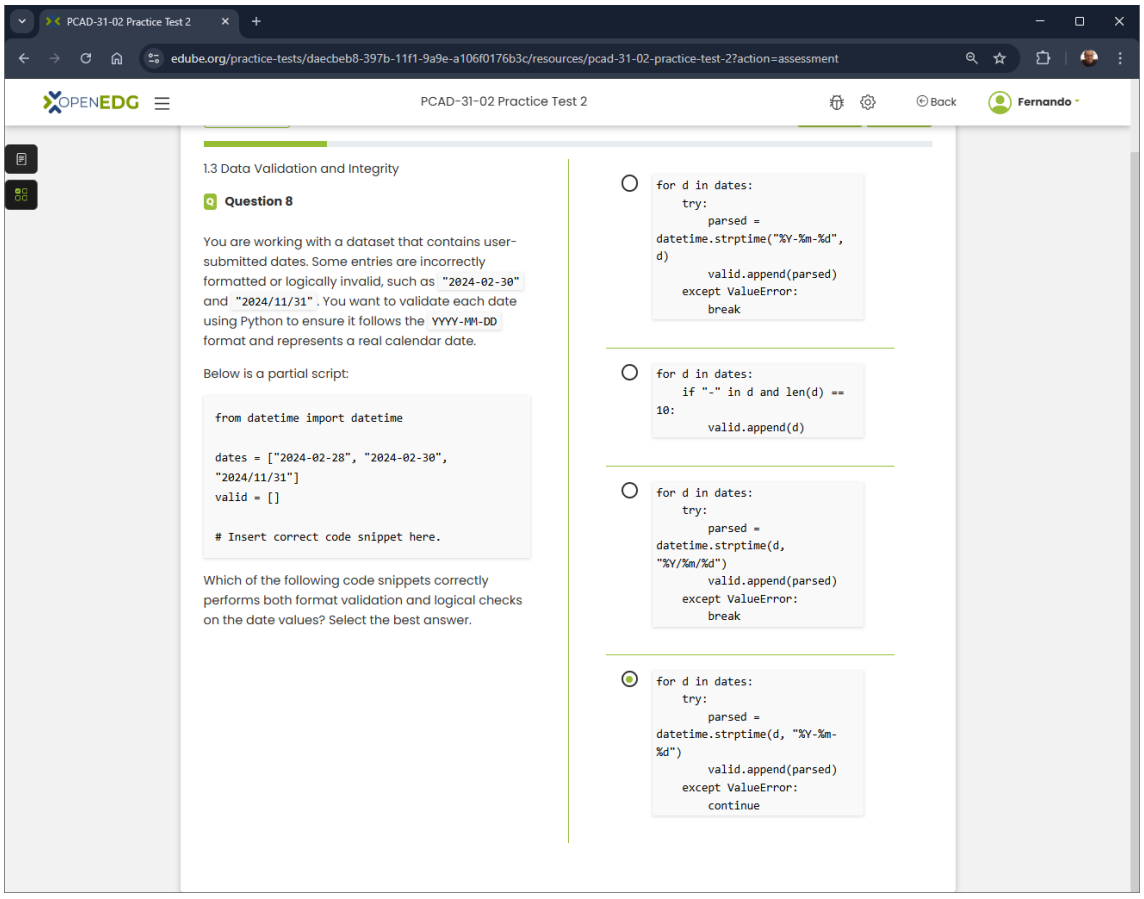

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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 3/48
Prev
Next
1.1 Data Collection, Integration, and Storage
Question 3
A data team is building a dashboard for a growing mobile app. The dashboard will pull daily usage stats from several tables, and must scale easily as more users join. The team wants high availability, built-in backups, and minimal maintenance effort.
Which storage solution best meets these needs? Select the best answer.
Use columnar data files stored in a remote folder to enable real-time updates and enforce transactional integrity.
Use a local file-based database, as it is lightweight and can be quickly set up for online applications.
Use XML files hosted on a remote server to simplify data access through standard parsers and reduce infrastructure complexity.
Use a managed cloud-based relational database that supports scalability, automated maintenance, and high availability.

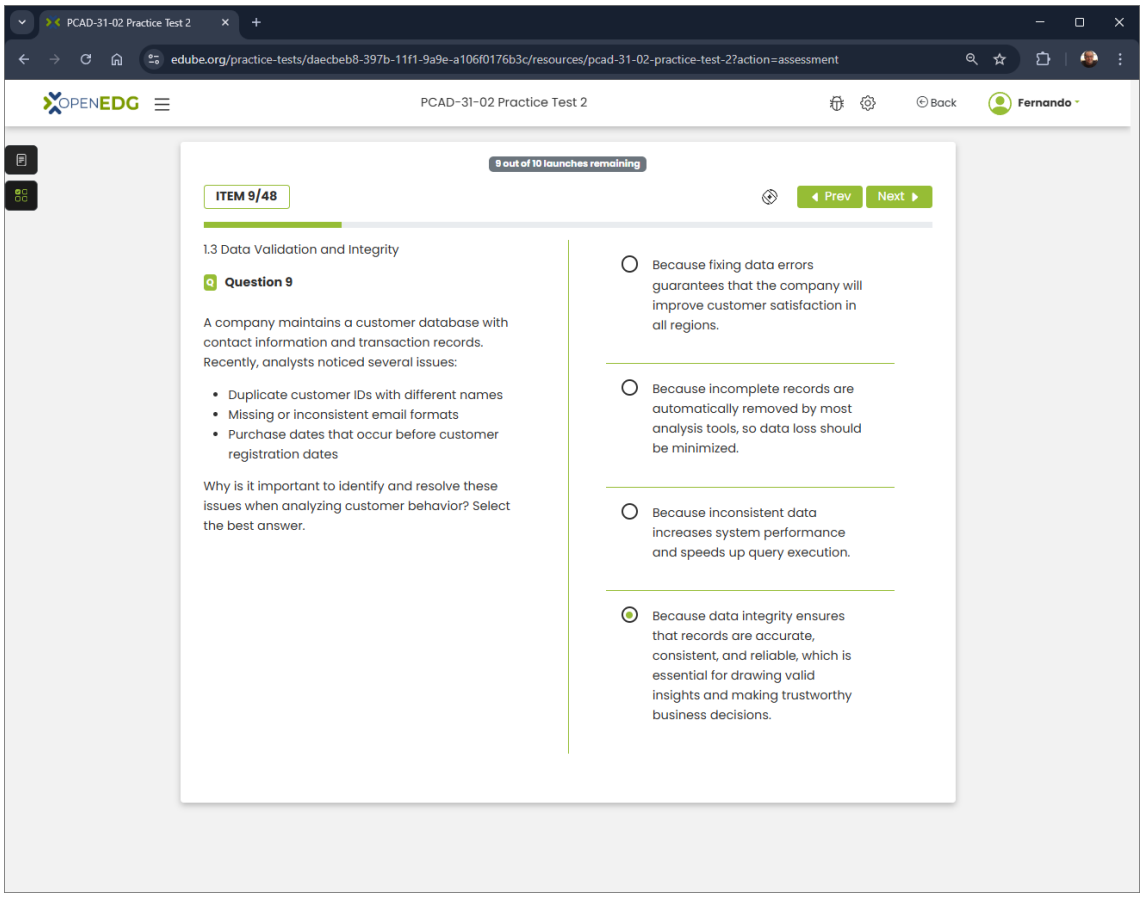

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 4/48
Prev
Next
1.2 Data Cleaning and Standardization
Question 4
A company is analyzing customer interactions from several sources: call center transcripts, chat logs, and a product catalog stored in a table. The analyst notes that the chat logs vary in length and contain emojis and abbreviations.
Which statement best explains the implications of working with this type of data? Select the best answer.
Structured data like the product catalog requires significantly more effort to clean and analyze than unstructured sources.
Data types do not affect analysis methods; all formats are processed the same once imported into a spreadsheet or database.
Unstructured data such as chat logs requires preprocessing and may involve text normalization, tokenization, and context-aware analysis to extract useful insights.
Unstructured data cannot be used in any analytical process and should be excluded unless converted into CSV format first.

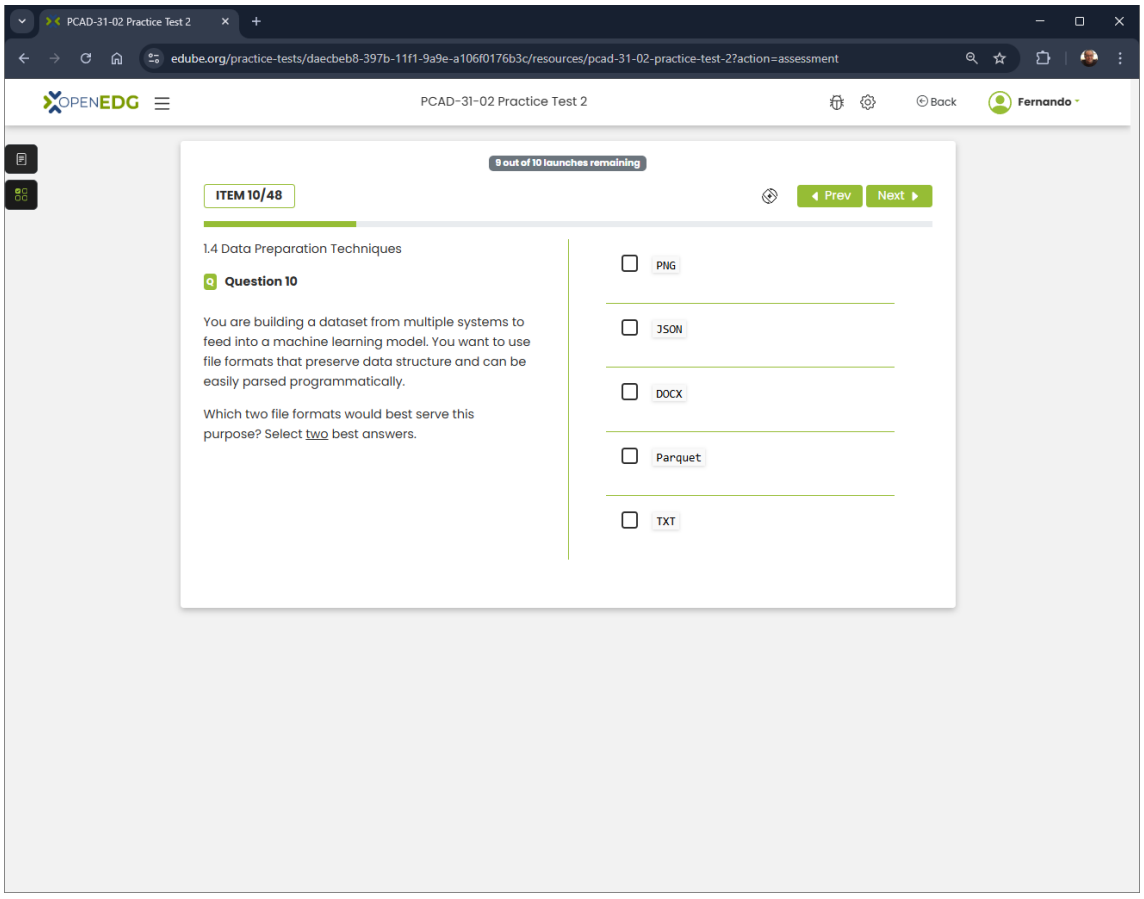

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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 5/48
Prev
Next
1.2 Data Cleaning and Standardization
Question 5
A data analyst reviews a survey dataset and notices the following responses in the "Age" field:
[28, 34, "twenty-nine", 52, -1, 125, 36, 75, 69]
Which approach is most appropriate before using this data for demographic analysis? Select the best answer.
Round all numeric ages to the nearest multiple of 5 for easy grouping in charts.
Standardize all valid entries to numeric form, flag or correct ambiguous text, and remove values that fall outside a plausible human age range.
Replace all entries with the average age to simplify the dataset and reduce variability.
Leave the data unchanged to avoid introducing bias during preprocessing.

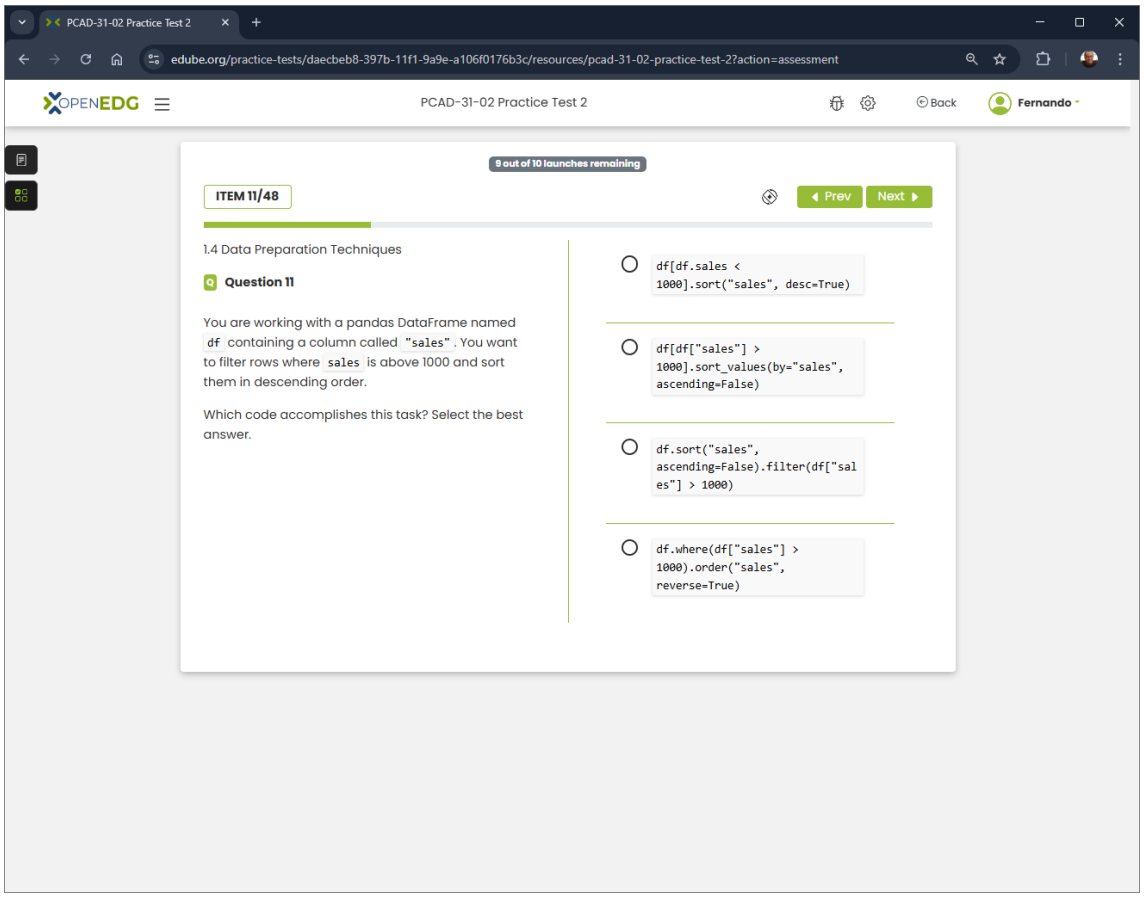

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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 6/48
Prev
Next
1.2 Data Cleaning and Standardization
Question 6
A dataset contains a column called "Department" with values like "Sales", "Support", and "Development". A data analyst needs to prepare this feature for use in a machine learning model.
Which approach enables the model to correctly interpret this categorical data? Select the best answer.
Use string length as a numeric proxy for each department name to retain variation.
Apply encoding techniques such as one-hot encoding or label encoding to convert the text values into numerical representations.
Replace each category with its average salary to simplify comparisons.
Remove the column to avoid introducing non-numeric values into the model.

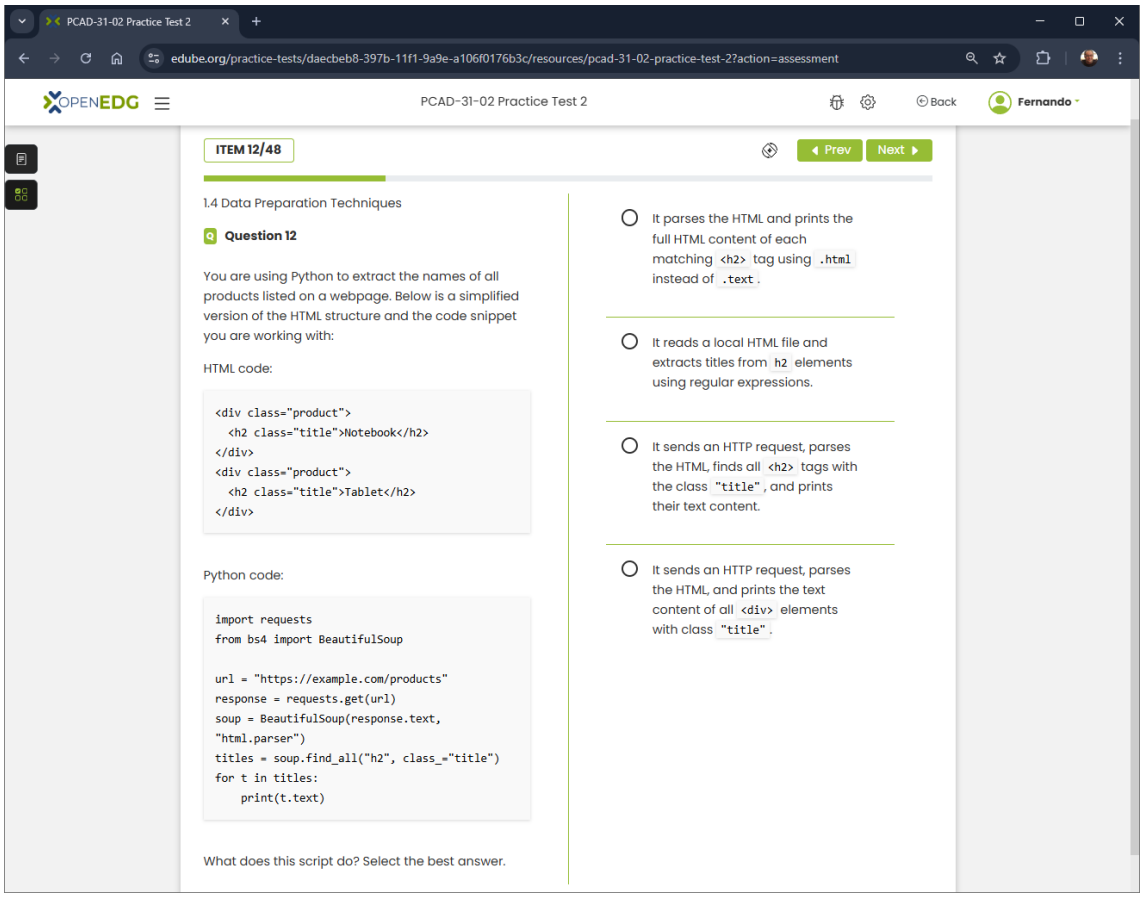

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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 7/48
Prev
Next
1.2 Data Cleaning and Standardization
Question 7
A data analyst is working with a dataset that includes a numeric column "Customer_Age". To simplify analysis and create clearer segments, the analyst converts the values into three groups: "18-30", "31-50", and "51+".
Which type of data transformation is the analyst applying? Select the best answer.
Data imputation – replacing missing values with estimated values.
Feature scaling – transforming values to a 0–1 range for comparison.
Label encoding – assigning numerical values to each unique category.
Bucketization – grouping continuous numeric values into predefined categorical ranges.


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
1.3 Data Validation and Integrity
Question 8
You are working with a dataset that contains user-submitted dates. Some entries are incorrectly formatted or logically invalid, such as "2024-02-30" and "2024/11/31". You want to validate each date using Python to ensure it follows the YYYY-MM-DD format and represents a real calendar date.
Below is a partial script:
from datetime import datetime
dates = ["2024-02-28", "2024-02-30", "2024/11/31"]
valid = []
Insert correct code snippet here.
Which of the following code snippets correctly performs both format validation and logical checks on the date values? Select the best answer.
for d in dates:
try:
parsed = datetime.strptime("%Y-%m-%d", d)
valid.append(parsed)
except ValueError:
break
for d in dates:
if "-" in d and len(d) == 10:
valid.append(d)
for d in dates:
try:
parsed = datetime.strptime(d, "%Y/%m/%d")
valid.append(parsed)
except ValueError:
break
for d in dates:
try:
parsed = datetime.strptime(d, "%Y-%m-%d")
valid.append(parsed)
except ValueError:
continue


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 9/48
Prev
Next
1.3 Data Validation and Integrity
Question 9
A company maintains a customer database with contact information and transaction records. Recently, analysts noticed several issues:
Duplicate customer IDs with different names
Missing or inconsistent email formats
Purchase dates that occur before customer registration dates
Why is it important to identify and resolve these issues when analyzing customer behavior? Select the best answer.
Because fixing data errors guarantees that the company will improve customer satisfaction in all regions.
Because incomplete records are automatically removed by most analysis tools, so data loss should be minimized.
Because inconsistent data increases system performance and speeds up query execution.
Because data integrity ensures that records are accurate, consistent, and reliable, which is essential for drawing valid insights and making trustworthy business decisions.


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 10/48
Prev
Next
1.4 Data Preparation Techniques
Question 10
You are building a dataset from multiple systems to feed into a machine learning model. You want to use file formats that preserve data structure and can be easily parsed programmatically.
Which two file formats would best serve this purpose? Select two best answers.
PNG
JSON
DOCX
Parquet
TXT

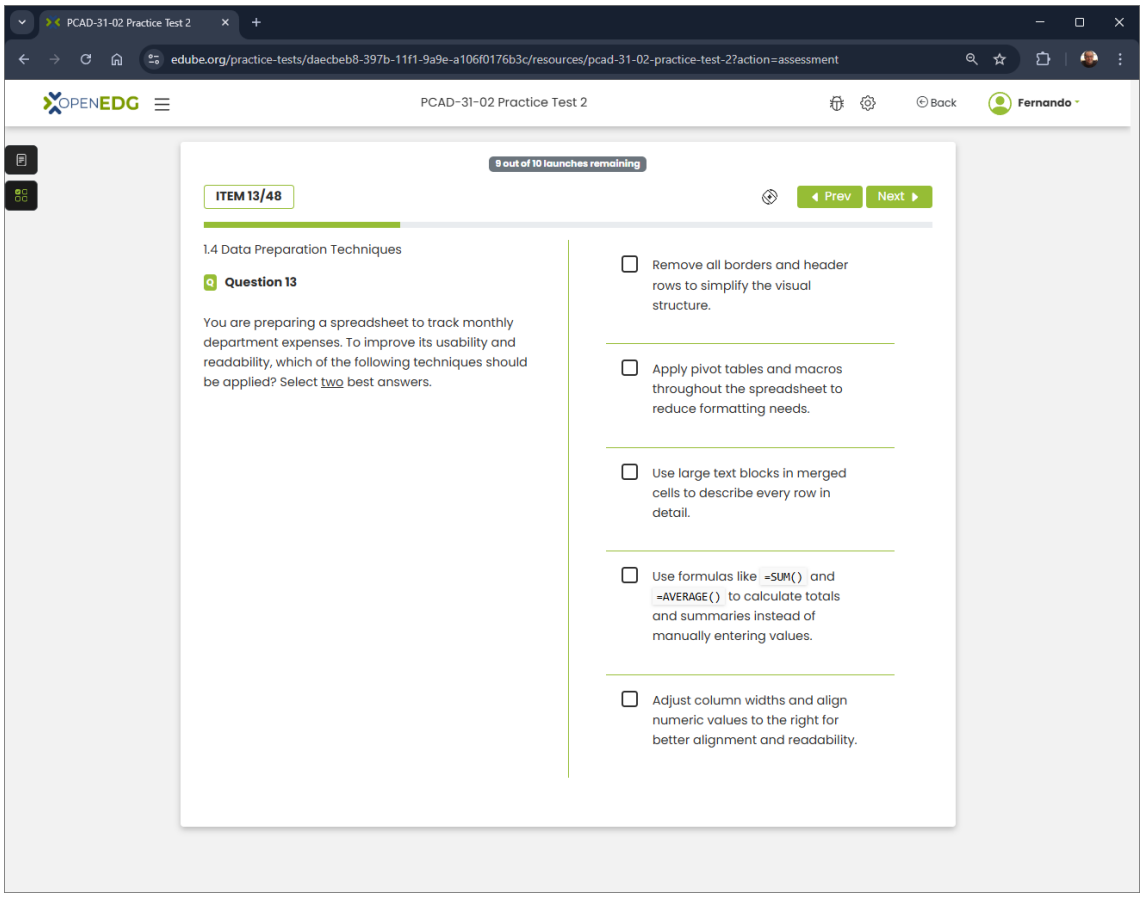

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 11/48
Prev
Next
1.4 Data Preparation Techniques
Question 11
You are working with a pandas DataFrame named df containing a column called "sales". You want to filter rows where sales is above 1000 and sort them in descending order.
Which code accomplishes this task? Select the best answer.
df[df.sales < 1000].sort("sales", desc=True)
df[df["sales"] > 1000].sort_values(by="sales", ascending=False)
df.sort("sales", ascending=False).filter(df["sales"] > 1000)
df.where(df["sales"] > 1000).order("sales", reverse=True)

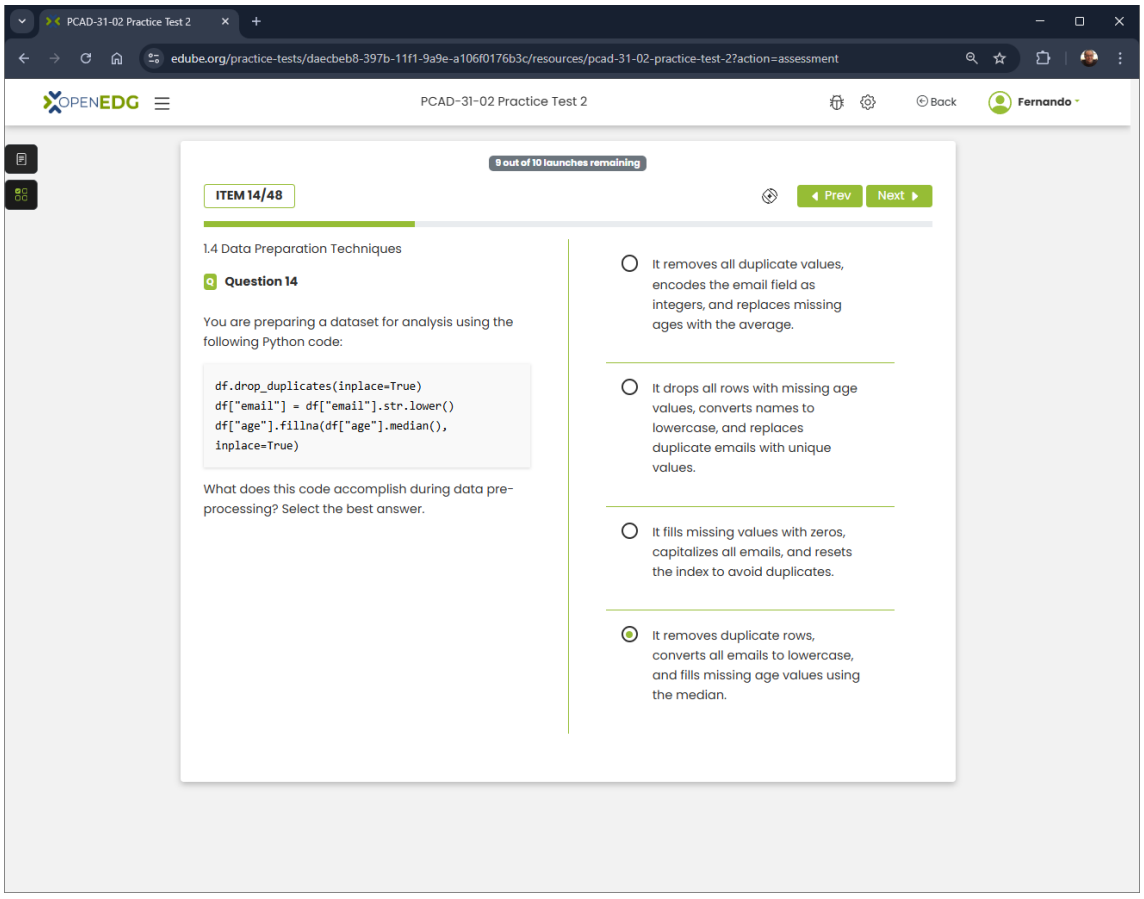

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
ITEM 12/48
Prev
Next
1.4 Data Preparation Techniques
Question 12
You are using Python to extract the names of all products listed on a webpage. Below is a simplified version of the HTML structure and the code snippet you are working with:
HTML code:

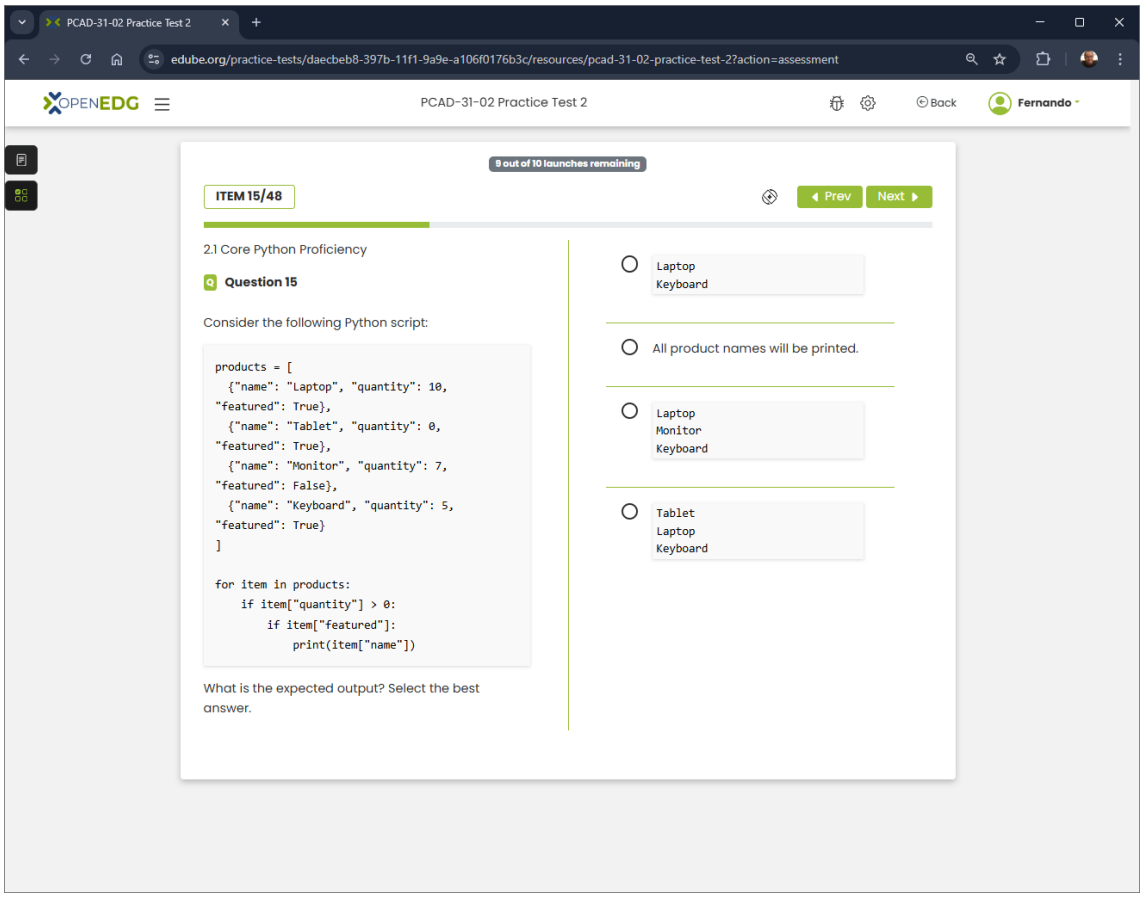
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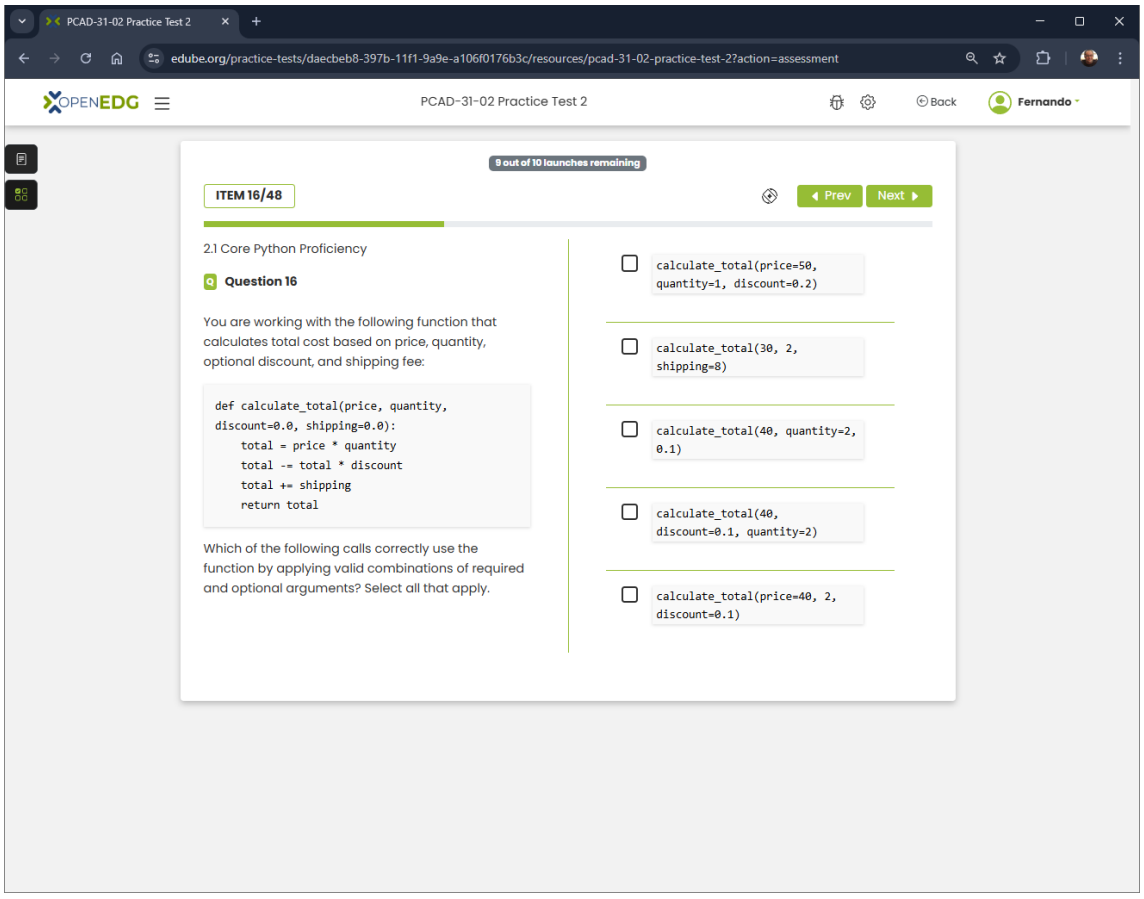
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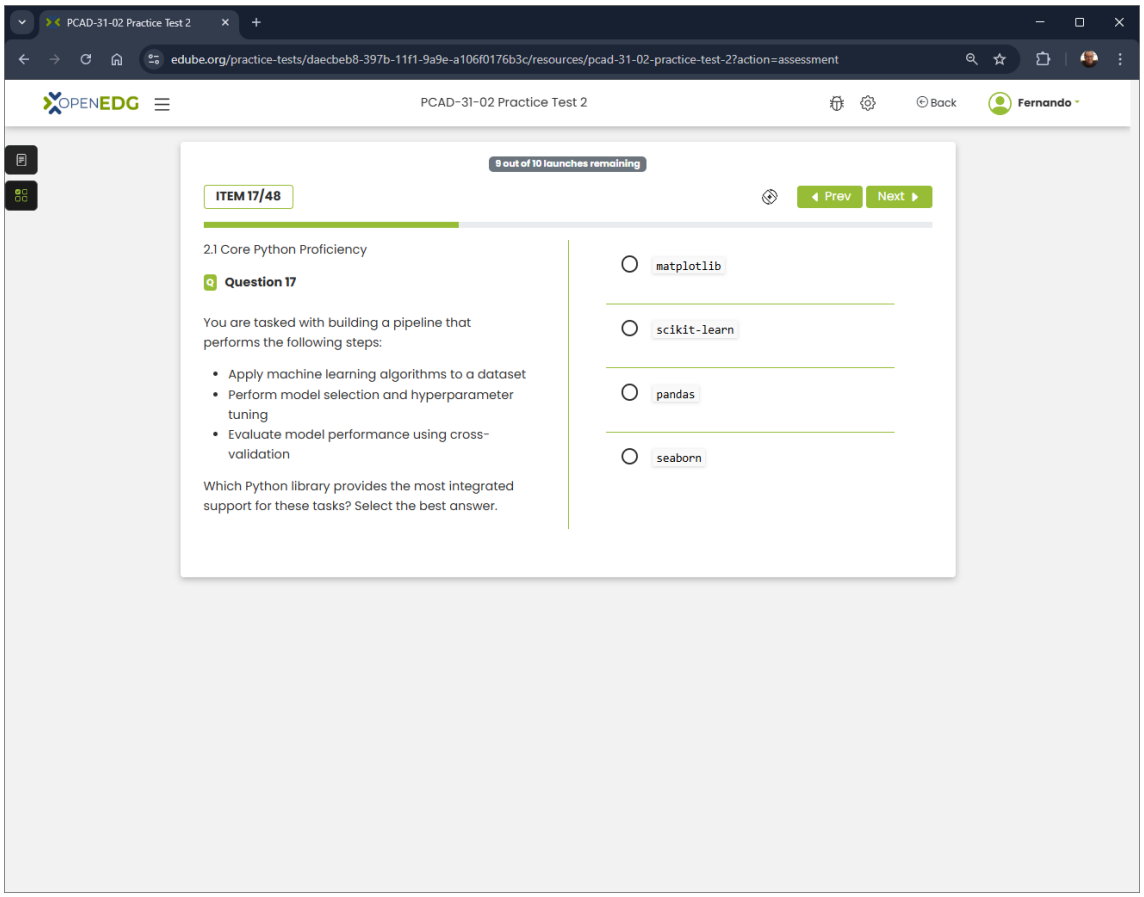
Python code:
import requests
from bs4 import BeautifulSoup
url = "https://example.com/products"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")
titles = soup.find_all("h2", class_="title")
for t in titles:
print(t.text)
What does this script do? Select the best answer.
It parses the HTML and prints the full HTML content of each matching <h2> tag using .html instead of .text.
It reads a local HTML file and extracts titles from h2 elements using regular expressions.
It sends an HTTP request, parses the HTML, finds all <h2> tags with the class "title", and prints their text content.
It sends an HTTP request, parses the HTML, and prints the text content of all elements with class "title".

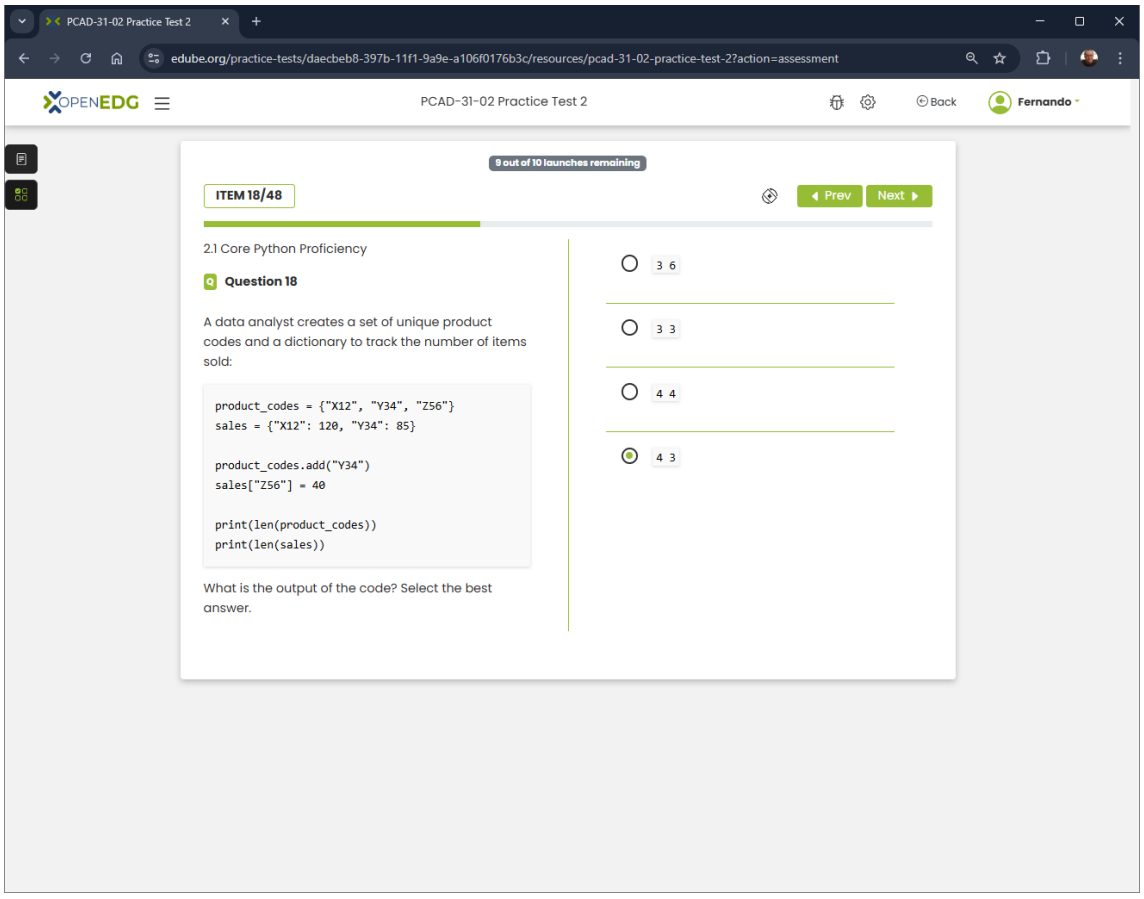

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 13/48
Prev
Next
1.4 Data Preparation Techniques
Question 13
You are preparing a spreadsheet to track monthly department expenses. To improve its usability and readability, which of the following techniques should be applied? Select two best answers.
Remove all borders and header rows to simplify the visual structure.
Apply pivot tables and macros throughout the spreadsheet to reduce formatting needs.
Use large text blocks in merged cells to describe every row in detail.
Use formulas like =SUM() and =AVERAGE() to calculate totals and summaries instead of manually entering values.
Adjust column widths and align numeric values to the right for better alignment and readability.

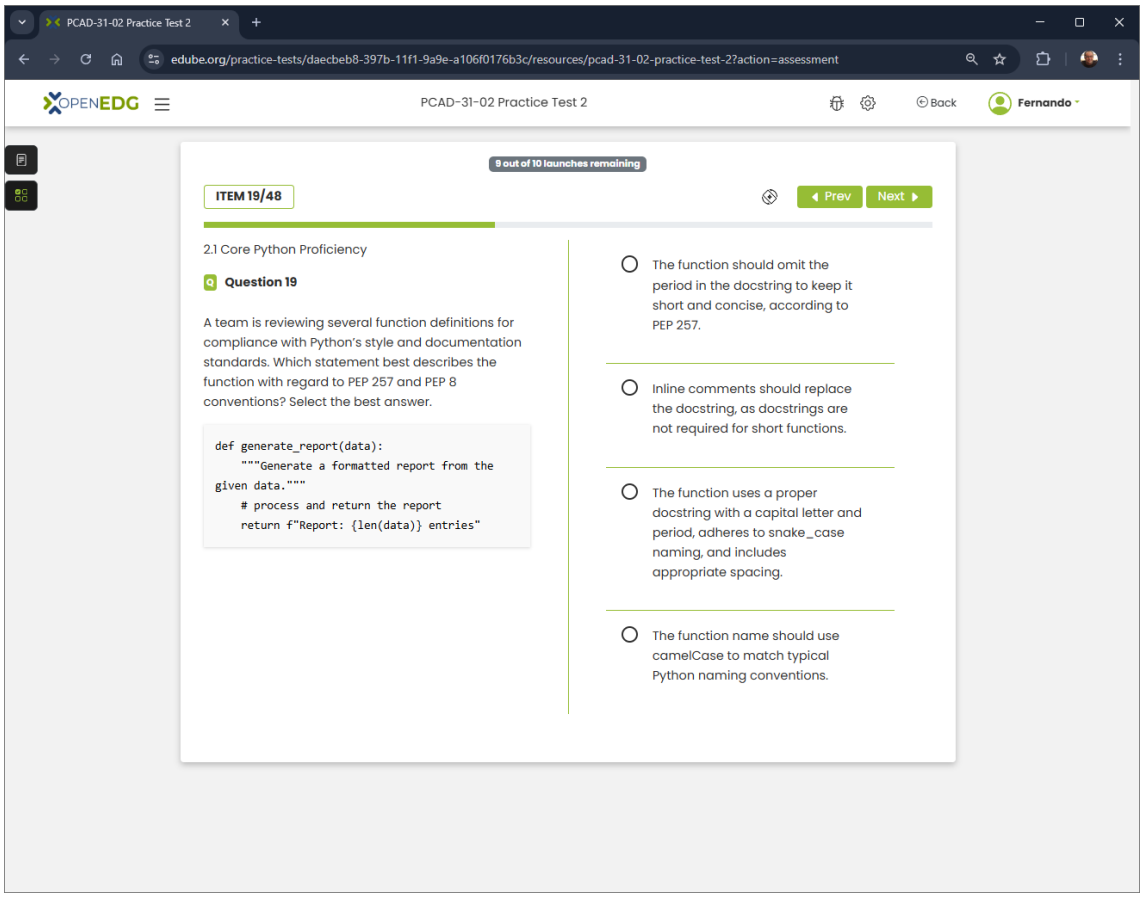

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 14/48
Prev
Next
1.4 Data Preparation Techniques
Question 14
You are preparing a dataset for analysis using the following Python code:
df.drop_duplicates(inplace=True)
df["email"] = df["email"].str.lower()
df["age"].fillna(df["age"].median(), inplace=True)
What does this code accomplish during data pre-processing? Select the best answer.
It removes all duplicate values, encodes the email field as integers, and replaces missing ages with the average.
It drops all rows with missing age values, converts names to lowercase, and replaces duplicate emails with unique values.
It fills missing values with zeros, capitalizes all emails, and resets the index to avoid duplicates.
It removes duplicate rows, converts all emails to lowercase, and fills missing age values using the median.

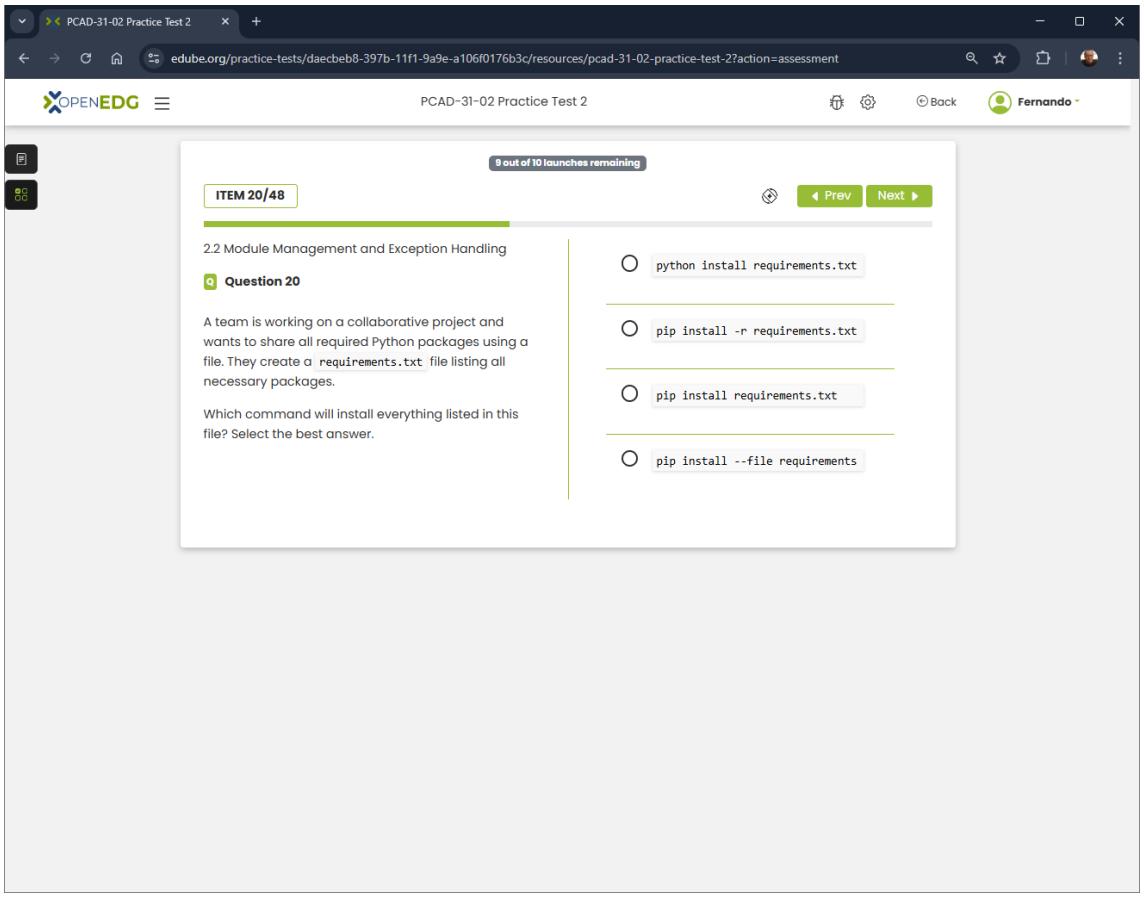

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 15/48
Prev
Next
2.1 Core Python Proficiency
Question 15
Consider the following Python script:
products = [
{"name": "Laptop", "quantity": 10,
"featured": True},
{"name": "Tablet", "quantity": 0,
"featured": True},
{"name": "Monitor", "quantity": 7,
"featured": False},
{"name": "Keyboard", "quantity": 5,
"featured": True}
]
for item in products:
if item["quantity"] > 0:
if item["featured"]:
print(item["name"])
What is the expected output? Select the best answer.
Laptop
Keyboard
All product names will be printed.
Laptop
Monitor
Keyboard
Tablet
Laptop
Keyboard

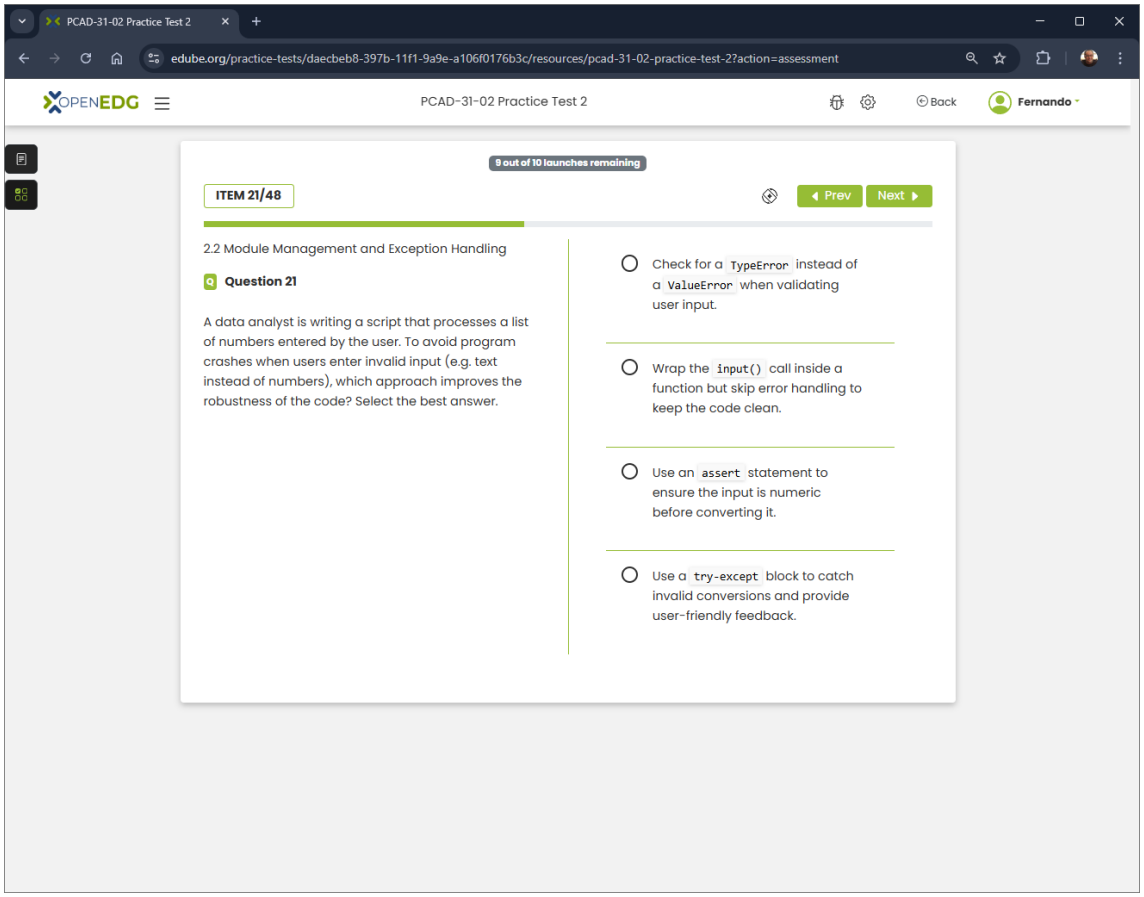

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 16/48
Prev
Next
2.1 Core Python Proficiency
Question 16
You are working with the following function that calculates total cost based on price, quantity, optional discount, and shipping fee:
def calculate_total(price, quantity,
discount=0.0, shipping=0.0):
total = price * quantity
total -= total * discount
total += shipping
return total
Which of the following calls correctly use the function by applying valid combinations of required and optional arguments? Select all that apply.
calculate_total(price=50,
quantity=1, discount=0.2)
calculate_total(30, 2,
shipping=8)
calculate_total(40, quantity=2,
0.1)
calculate_total(40,
discount=0.1, quantity=2)
calculate_total(price=40, 2,
discount=0.1)

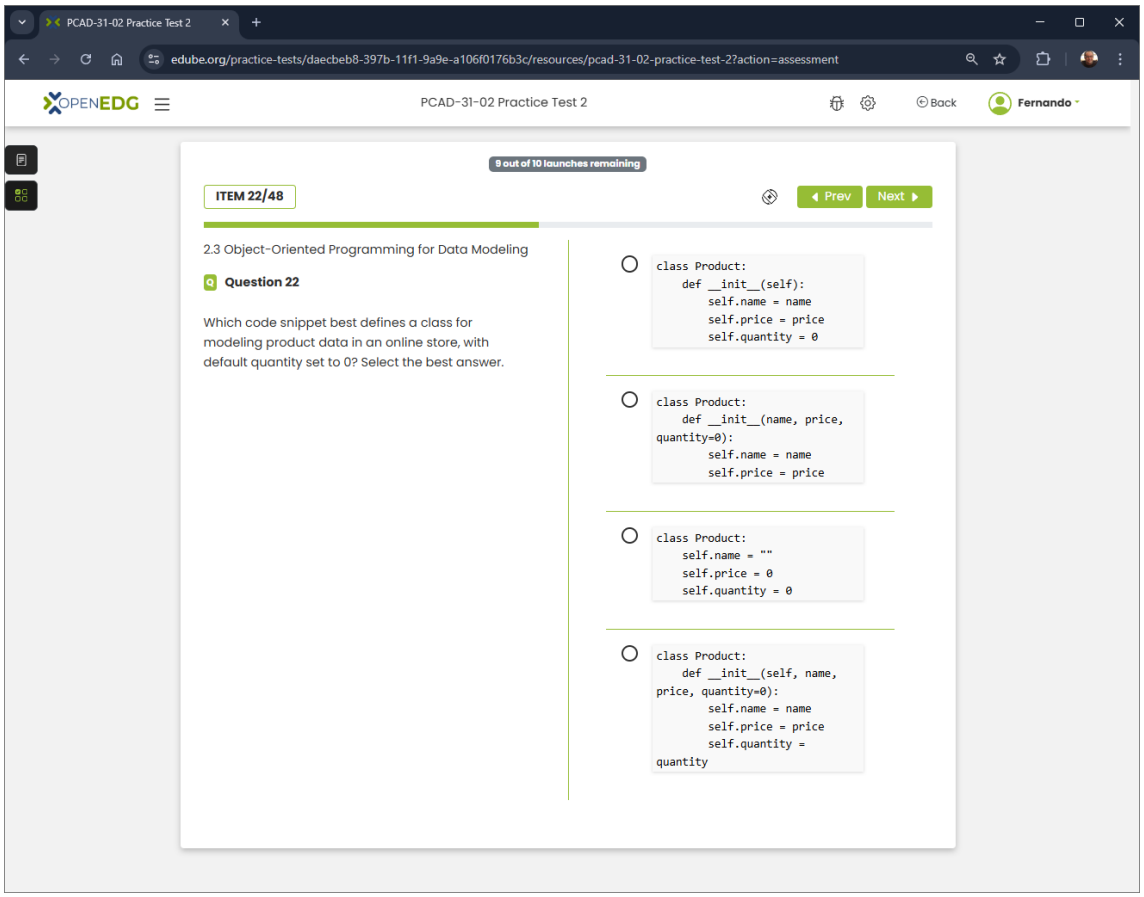

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 17/48
Prev
Next
2.1 Core Python Proficiency
Question 17
You are tasked with building a pipeline that performs the following steps:
Apply machine learning algorithms to a dataset
Perform model selection and hyperparameter tuning
Evaluate model performance using cross-validation
Which Python library provides the most integrated support for these tasks? Select the best answer.
matplotlib
scikit-learn
pandas
seaborn

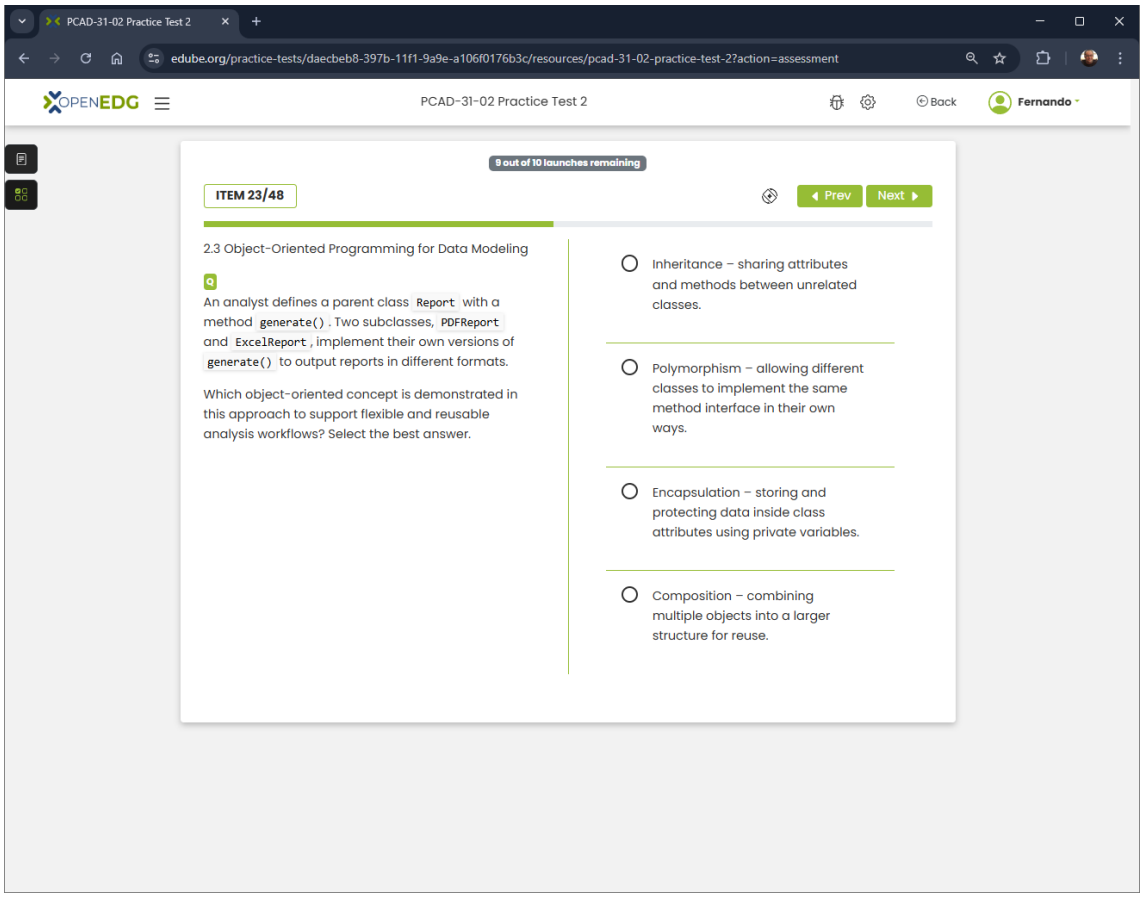

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 18/48
Prev
Next
2.1 Core Python Proficiency
Question 18
A data analyst creates a set of unique product codes and a dictionary to track the number of items sold:
product_codes = {"X12", "Y34", "Z56"}
sales = {"X12": 120, "Y34": 85}
product_codes.add("Y34")
sales["Z56"] = 40
print(len(product_codes))
print(len(sales))
What is the output of the code? Select the best answer.
3 6
3 3
4 4
4 3

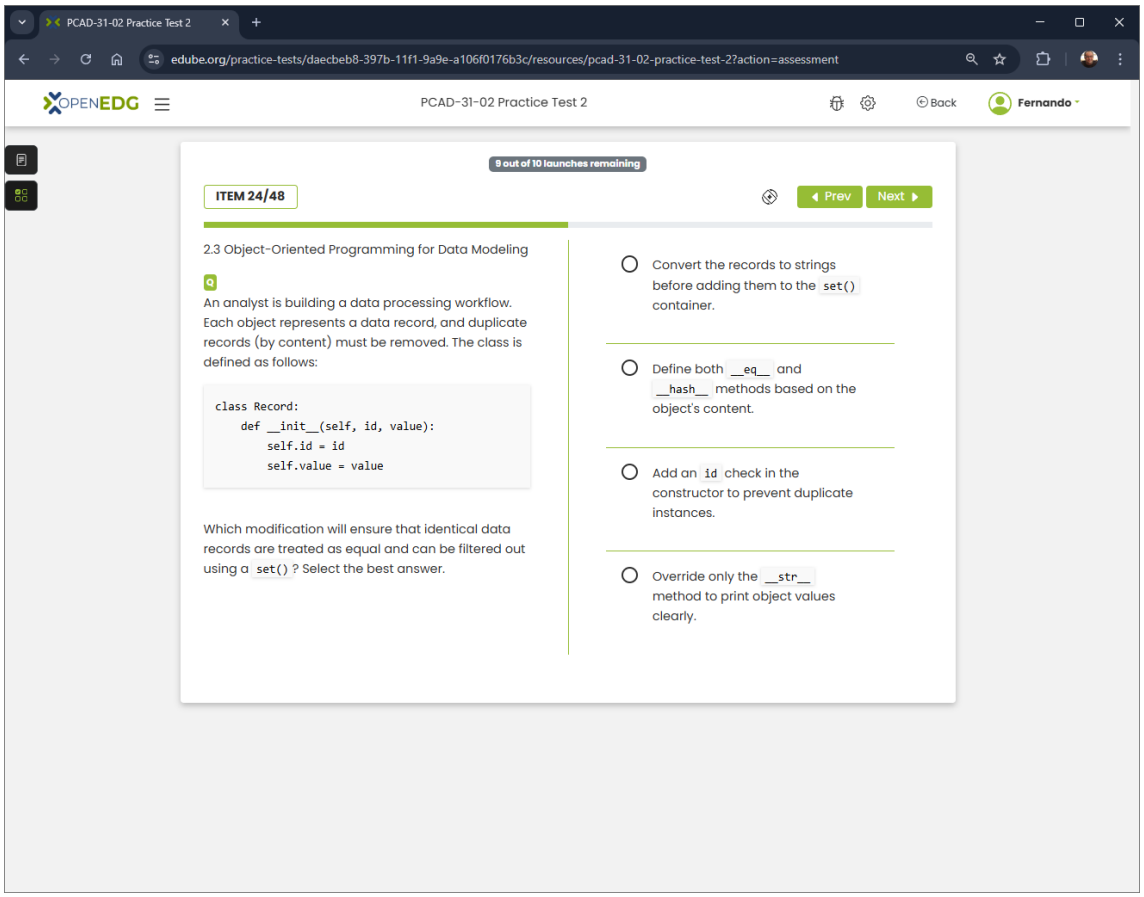

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 19/48
Prev
Next
2.1 Core Python Proficiency
Question 19
A team is reviewing several function definitions for compliance with Python's style and documentation standards. Which statement best describes the function with regard to PEP 257 and PEP 8 conventions? Select the best answer.
def generate_report(data):
"""Generate a formatted report from the given data."""
process and return the report
return f"Report: {len(data)} entries"
The function should omit the period in the docstring to keep it short and concise, according to PEP 257.
Inline comments should replace the docstring, as docstrings are not required for short functions.
The function uses a proper docstring with a capital letter and period, adheres to snake_case naming, and includes appropriate spacing.
The function name should use camelCase to match typical Python naming conventions.

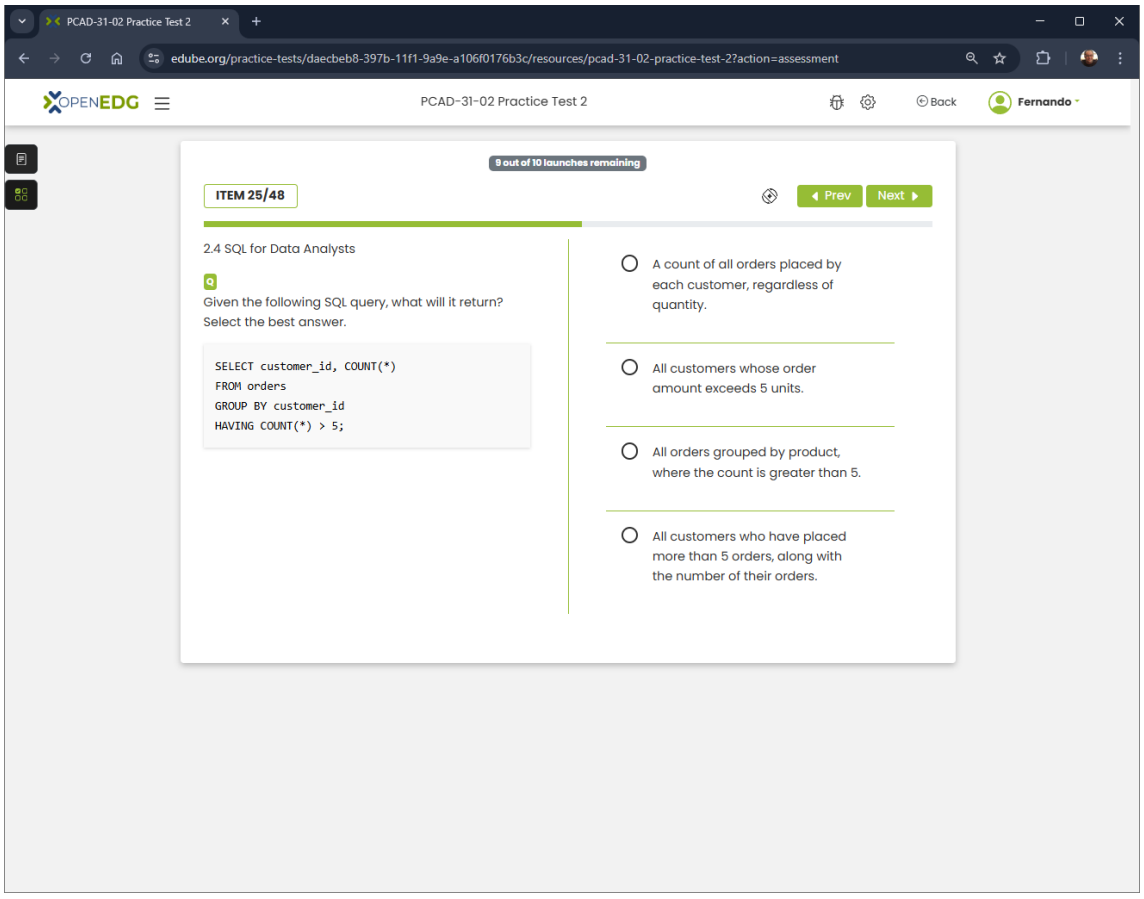

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 20/48
Prev
Next
2.2 Module Management and Exception Handling
Question 20
A team is working on a collaborative project and wants to share all required Python packages using a file. They create a requirements.txt file listing all necessary packages.
Which command will install everything listed in this file? Select the best answer.
python install requirements.txt
pip install -r requirements.txt
pip install requirements.txt
pip install --file requirements

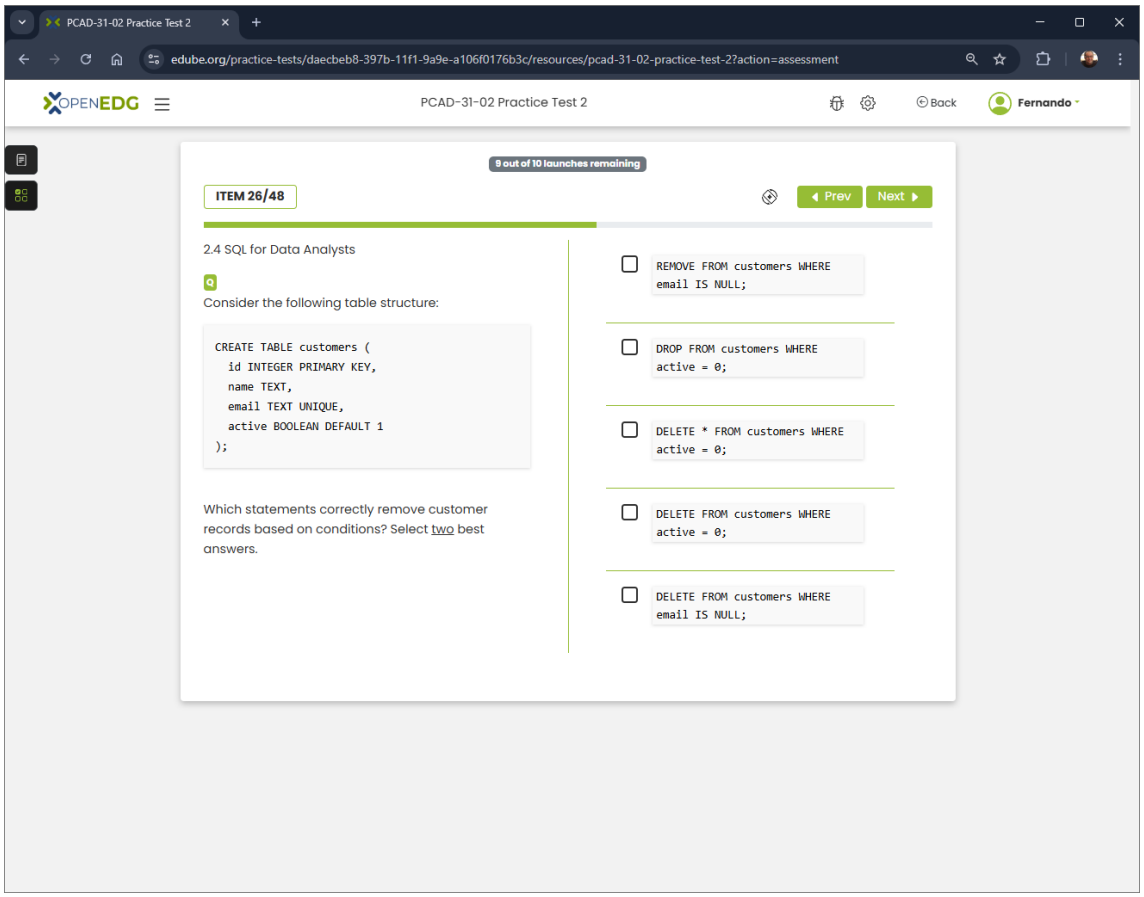

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 21/48
Prev
Next
2.2 Module Management and Exception Handling
Question 21
A data analyst is writing a script that processes a list of numbers entered by the user. To avoid program crashes when users enter invalid input (e.g. text instead of numbers), which approach improves the robustness of the code? Select the best answer.
Check for a TypeError instead of a ValueError when validating user input.
Wrap the input() call inside a function but skip error handling to keep the code clean.
Use an assert statement to ensure the input is numeric before converting it.
Use a try-except block to catch invalid conversions and provide user-friendly feedback.

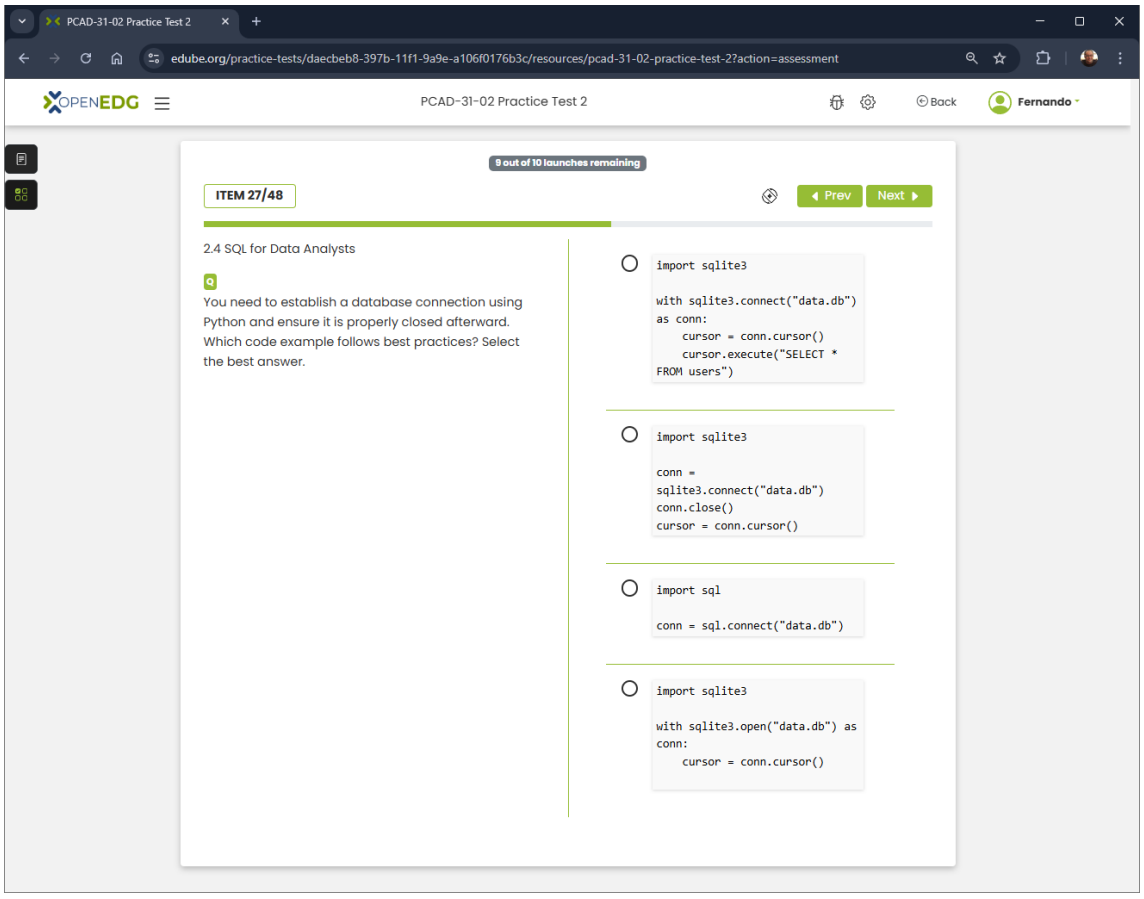

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 22/48
Prev
Next
2.3 Object-Oriented Programming for Data Modeling
Question 22
Which code snippet best defines a class for modeling product data in an online store, with default quantity set to 0? Select the best answer.
class Product:
def __init__(self):
self.name = name
self.price = price
self.quantity = 0
class Product:
def __init__(name, price, quantity=0):
self.name = name
self.price = price
class Product:
self.name = ""
self.price = 0
self.quantity = 0
class Product:
def __init__(self, name, price, quantity=0):
self.name = name
self.price = price
self.quantity = quantity

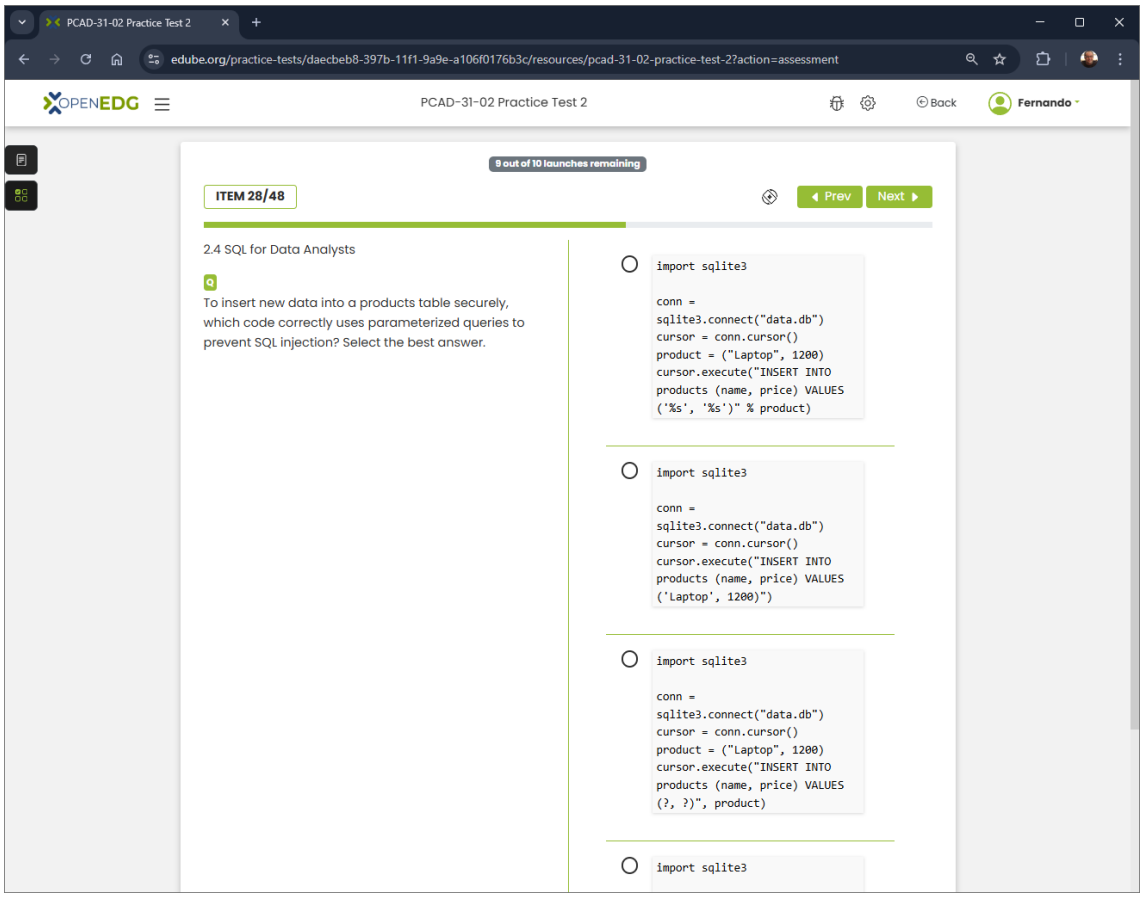

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 23/48
Prev
Next
2.3 Object-Oriented Programming for Data Modeling
An analyst defines a parent class Report with a method generate(). Two subclasses, PDFReport and ExcelReport, implement their own versions of generate() to output reports in different formats.
Which object-oriented concept is demonstrated in this approach to support flexible and reusable analysis workflows? Select the best answer.
Inheritance – sharing attributes and methods between unrelated classes.
Polymorphism – allowing different classes to implement the same method interface in their own ways.
Encapsulation – storing and protecting data inside class attributes using private variables.
Composition – combining multiple objects into a larger structure for reuse.

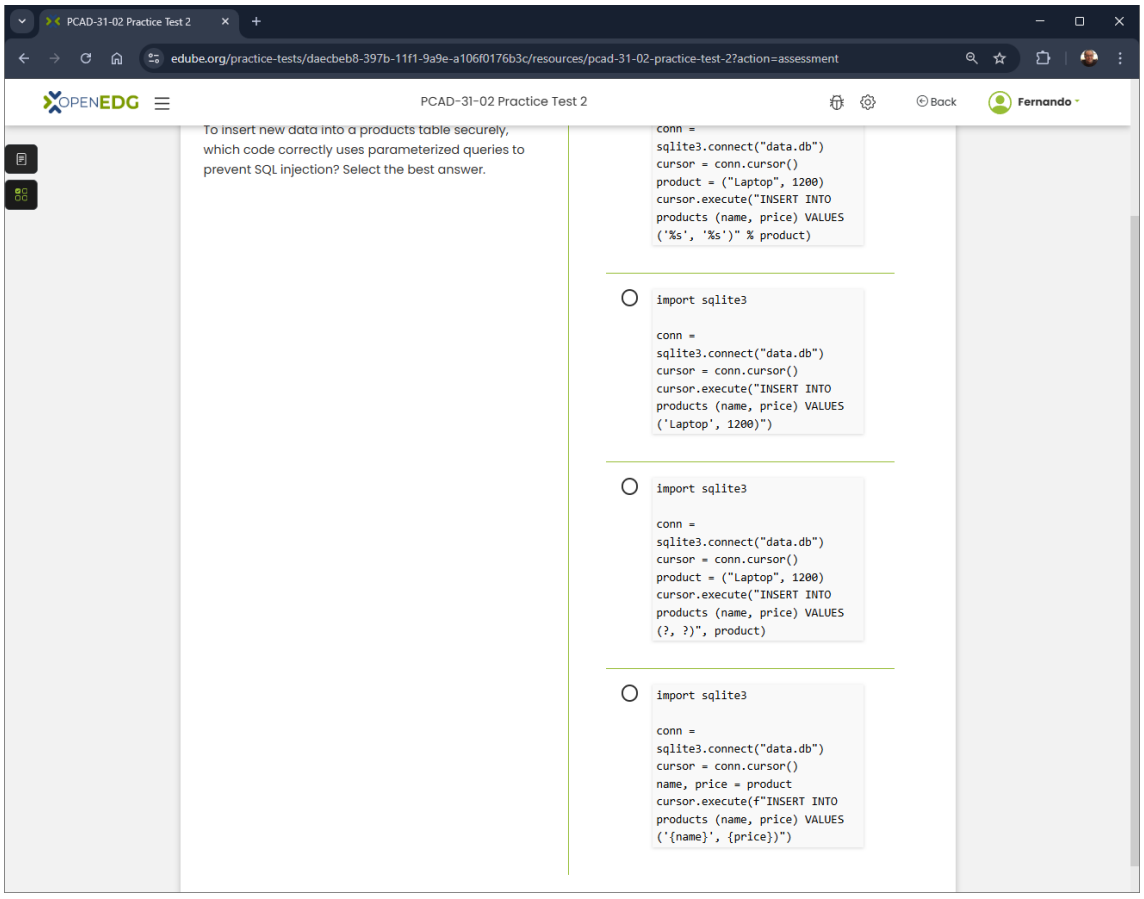

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 24/48
Prev
Next
2.3 Object-Oriented Programming for Data Modeling
An analyst is building a data processing workflow. Each object represents a data record, and duplicate records (by content) must be removed. The class is defined as follows:
class Record:
def __init__(self, id, value):
self.id = id
self.value = value
Which modification will ensure that identical data records are treated as equal and can be filtered out using a set() ? Select the best answer.
Convert the records to strings before adding them to the set() container.
Define both __eq__ and __hash__ methods based on the object's content.
Add an id check in the constructor to prevent duplicate instances.
Override only the __str__ method to print object values clearly.

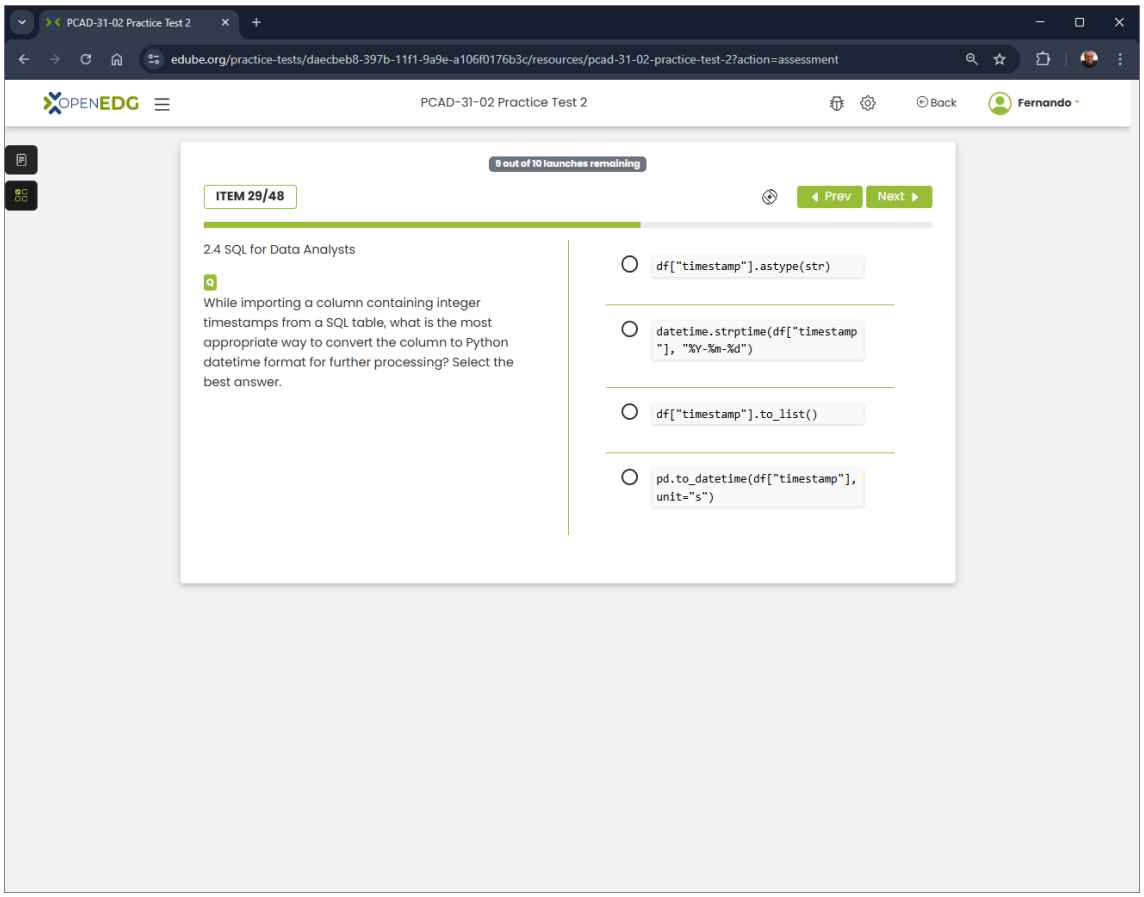

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 25/48
Prev
Next
2.4 SQL for Data Analysts
Given the following SQL query, what will it return? Select the best answer.
SELECT customer_id, COUNT(*)
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 5;
A count of all orders placed by each customer, regardless of quantity.
All customers whose order amount exceeds 5 units.
All orders grouped by product, where the count is greater than 5.
All customers who have placed more than 5 orders, along with the number of their orders.

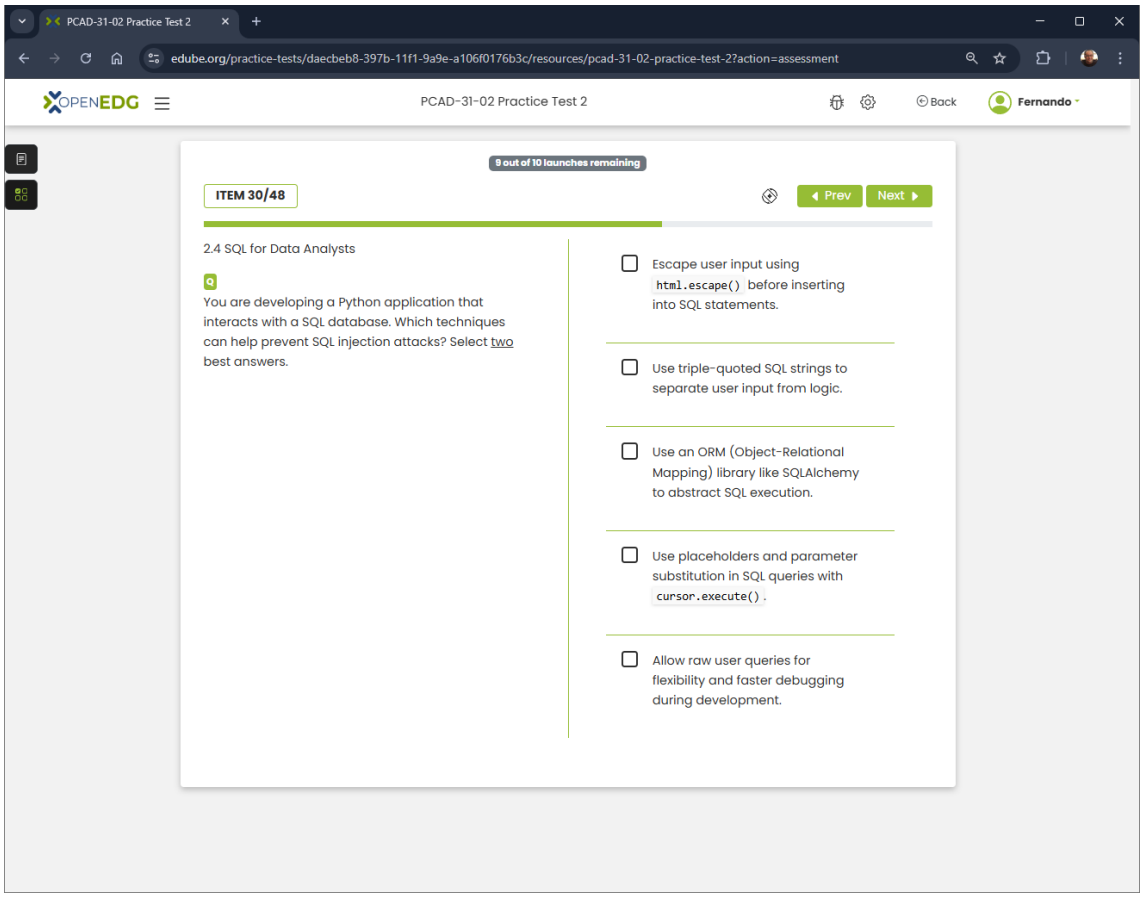

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 26/48
Prev
Next
2.4 SQL for Data Analysts
Consider the following table structure:
CREATE TABLE customers (
id INTEGER PRIMARY KEY,
name TEXT,
email TEXT UNIQUE,
active BOOLEAN DEFAULT 1
);
Which statements correctly remove customer records based on conditions? Select two best answers.
REMOVE FROM customers WHERE email IS NULL;
DROP FROM customers WHERE active = 0;
DELETE * FROM customers WHERE active = 0;
DELETE FROM customers WHERE active = 0;
DELETE FROM customers WHERE email IS NULL;


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 27/48
Prev
Next
2.4 SQL for Data Analysts
You need to establish a database connection using Python and ensure it is properly closed afterward. Which code example follows best practices? Select the best answer.
import sqlite3
with sqlite3.connect("data.db") as conn:
cursor = conn.cursor()
cursor.execute("SELECT * FROM users")
import sqlite3
conn = sqlite3.connect("data.db")
conn.close()
cursor = conn.cursor()
import sql
conn = sql.connect("data.db")
import sqlite3
with sqlite3.open("data.db") as conn:
cursor = conn.cursor()


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 28/48
Prev
Next
2.4 SQL for Data Analysts
Q
To insert new data into a products table securely, which code correctly uses parameterized queries to prevent SQL injection? Select the best answer.
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
product = ("Laptop", 1200)
cursor.execute("INSERT INTO products (name, price) VALUES ('%s', '%s')" % product)
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
cursor.execute("INSERT INTO products (name, price) VALUES ('Laptop', 1200)")
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
product = ("Laptop", 1200)
cursor.execute("INSERT INTO products (name, price) VALUES (?, ?)", product)
import sqlite3


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
To insert new data into a products table securely, which code correctly uses parameterized queries to prevent SQL injection? Select the best answer.
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
product = ("Laptop", 1200)
cursor.execute("INSERT INTO products (name, price) VALUES ('%s', '%s')" % product)
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
cursor.execute("INSERT INTO products (name, price) VALUES ('Laptop', 1200)")
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
product = ("Laptop", 1200)
cursor.execute("INSERT INTO products (name, price) VALUES (?, ?)", product)
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
name, price = product
cursor.execute(f"INSERT INTO products (name, price) VALUES ('{name}', {price})")


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 29/48
Prev
Next
2.4 SQL for Data Analysts
While importing a column containing integer timestamps from a SQL table, what is the most appropriate way to convert the column to Python datetime format for further processing? Select the best answer.
df["timestamp"].astype(str)
datetime.strptime(df["timestamp"], "%Y-%m-%d")
df["timestamp"].to_list()
pd.to_datetime(df["timestamp"], unit="s")


PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 30/48
Prev
Next
2.4 SQL for Data Analysts
You are developing a Python application that interacts with a SQL database. Which techniques can help prevent SQL injection attacks? Select two best answers.
Escape user input using html.escape() before inserting into SQL statements.
Use triple-quoted SQL strings to separate user input from logic.
Use an ORM (Object-Relational Mapping) library like SQLAlchemy to abstract SQL execution.
Use placeholders and parameter substitution in SQL queries with cursor.execute().
Allow raw user queries for flexibility and faster debugging during development.

PCAD-31-02 Practice Test 2

edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment

OPENEDG

PCAD-31-02 Practice Test 2

Back

Fernando

9 out of 10 launches remaining

ITEM 31/48

Prev Next

3.1 Descriptive Statistics

A dataset displays a symmetric, bell-shaped distribution. A data analyst reports a 95% confidence interval for the mean.

Which of the following statements are most likely true? Select two best answers.

- [ ] The mean, median, and mode are approximately equal.
- [ ] The distribution is skewed to the right, with most values below the mean.
- [ ] About 95% of the data values lie within 2 standard deviations of the mean.
- [ ] The standard deviation in a bell-shaped curve is always below the mean, confirming that the data is tightly clustered.
- [ ] A 95% confidence interval guarantees that 95% of future values will fall within the specified range.

PCAD-31-02 Practice Test 2

edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment

OPENEDG

PCAD-31-02 Practice Test 2

Back

Fernando

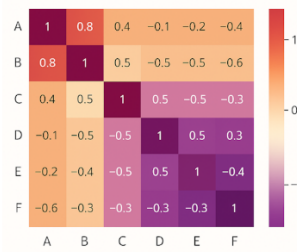
9 out of 10 launches remaining

ITEM 32/48

Prev Next

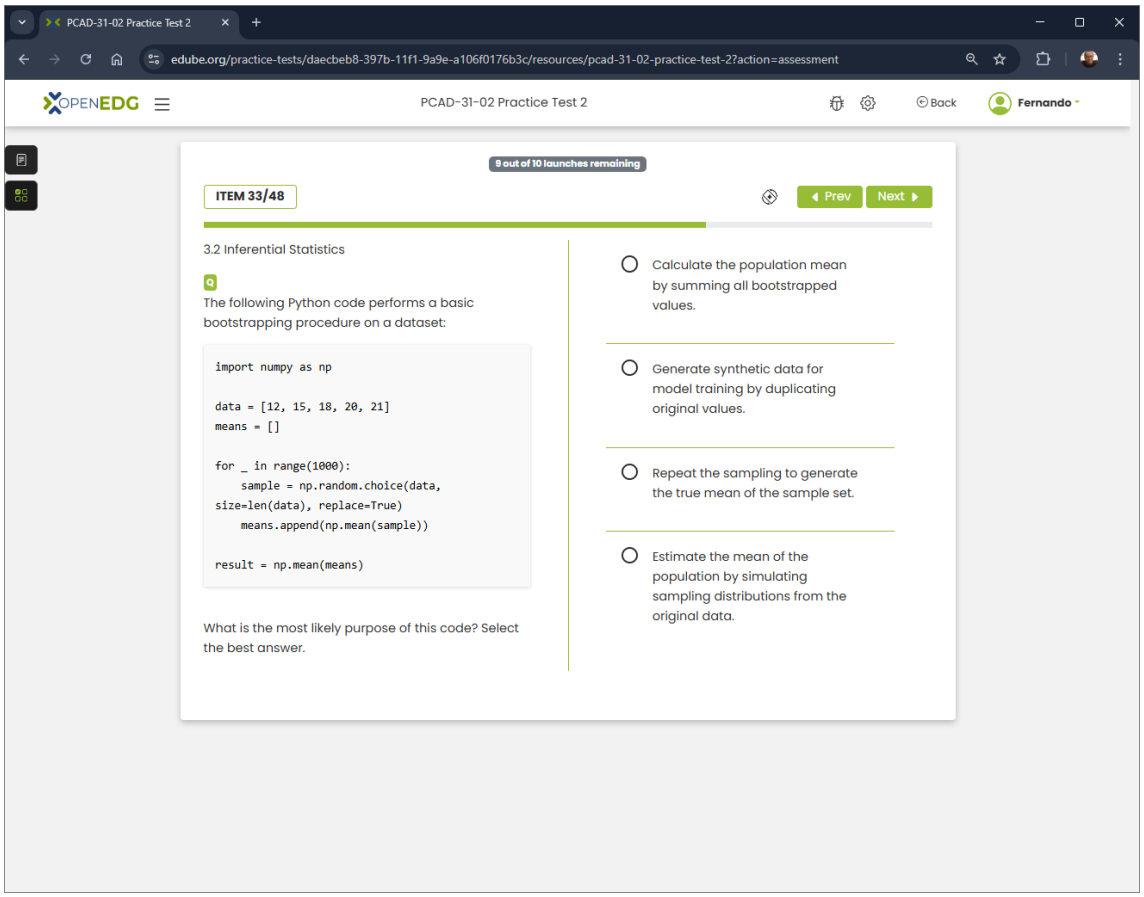
3.1 Descriptive Statistics

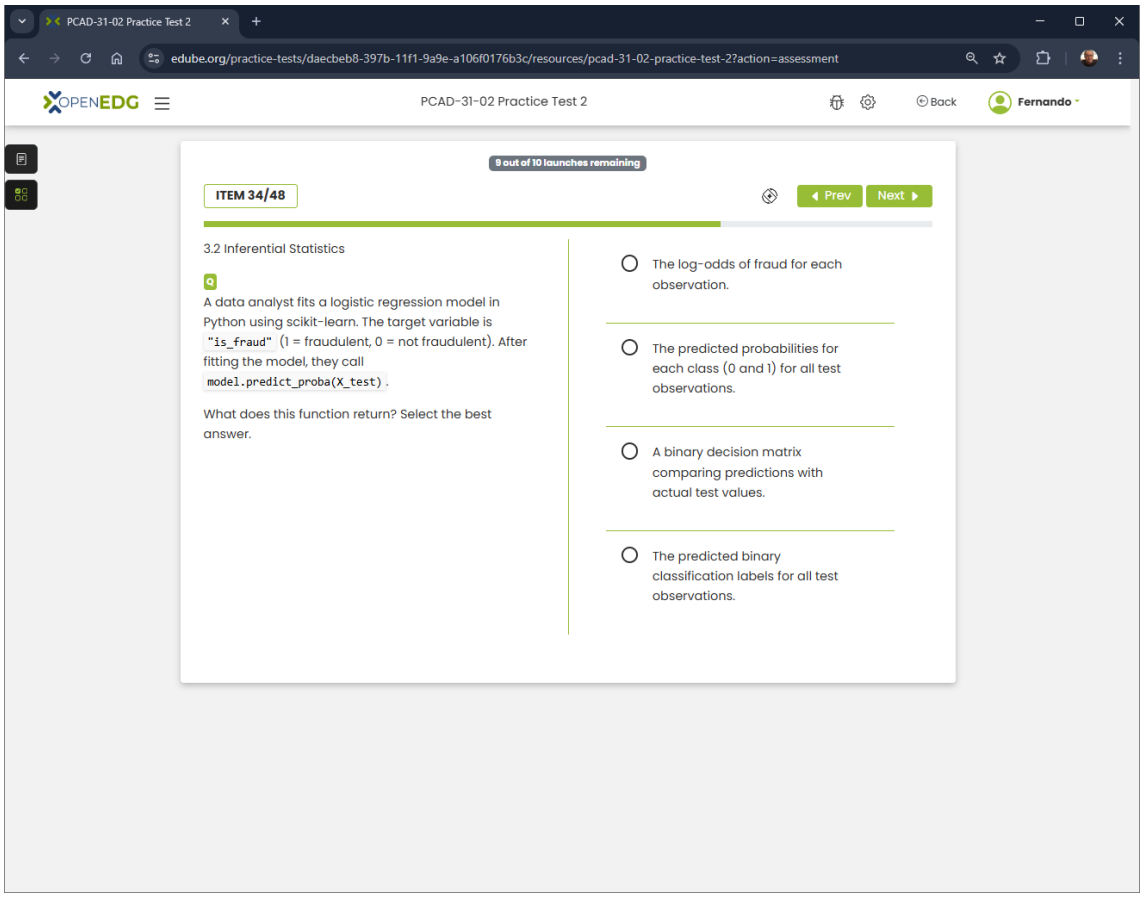
A data analyst creates the following correlation heatmap to explore relationships between variables in a dataset:

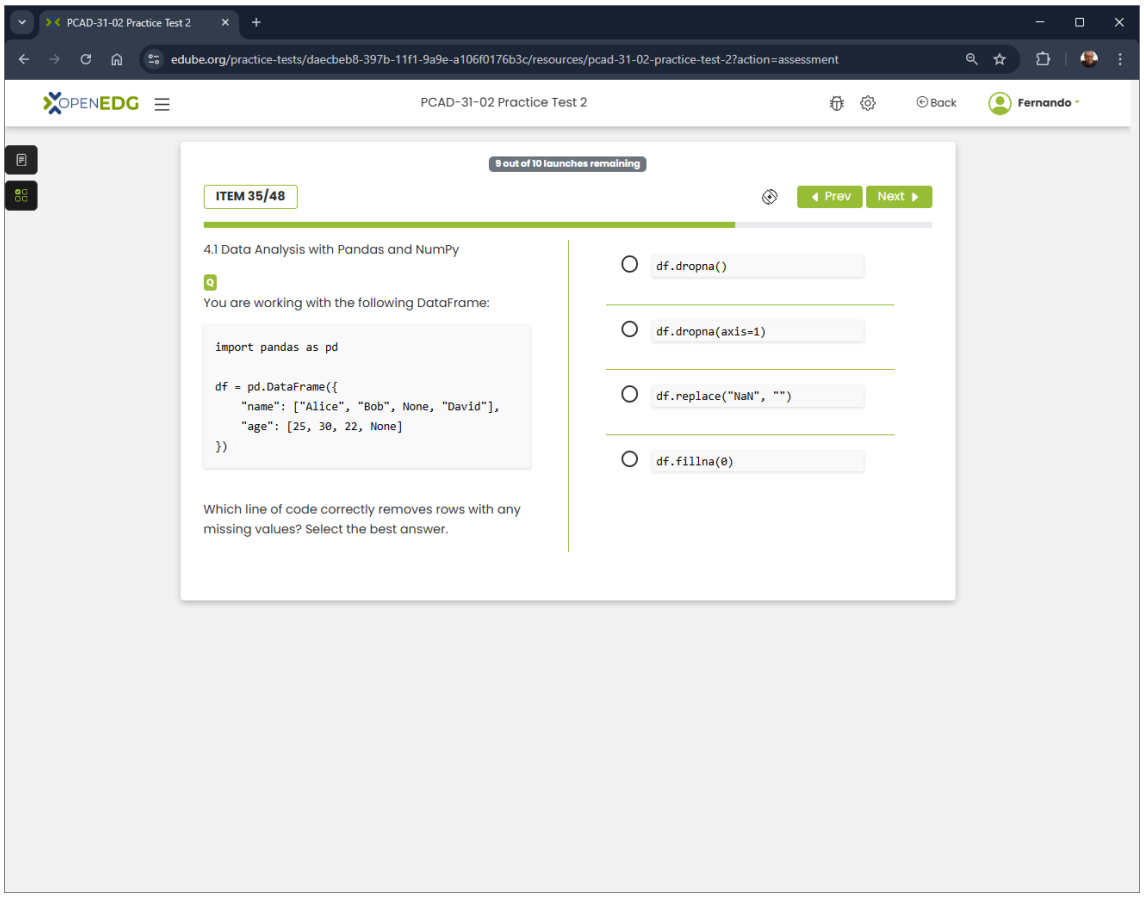


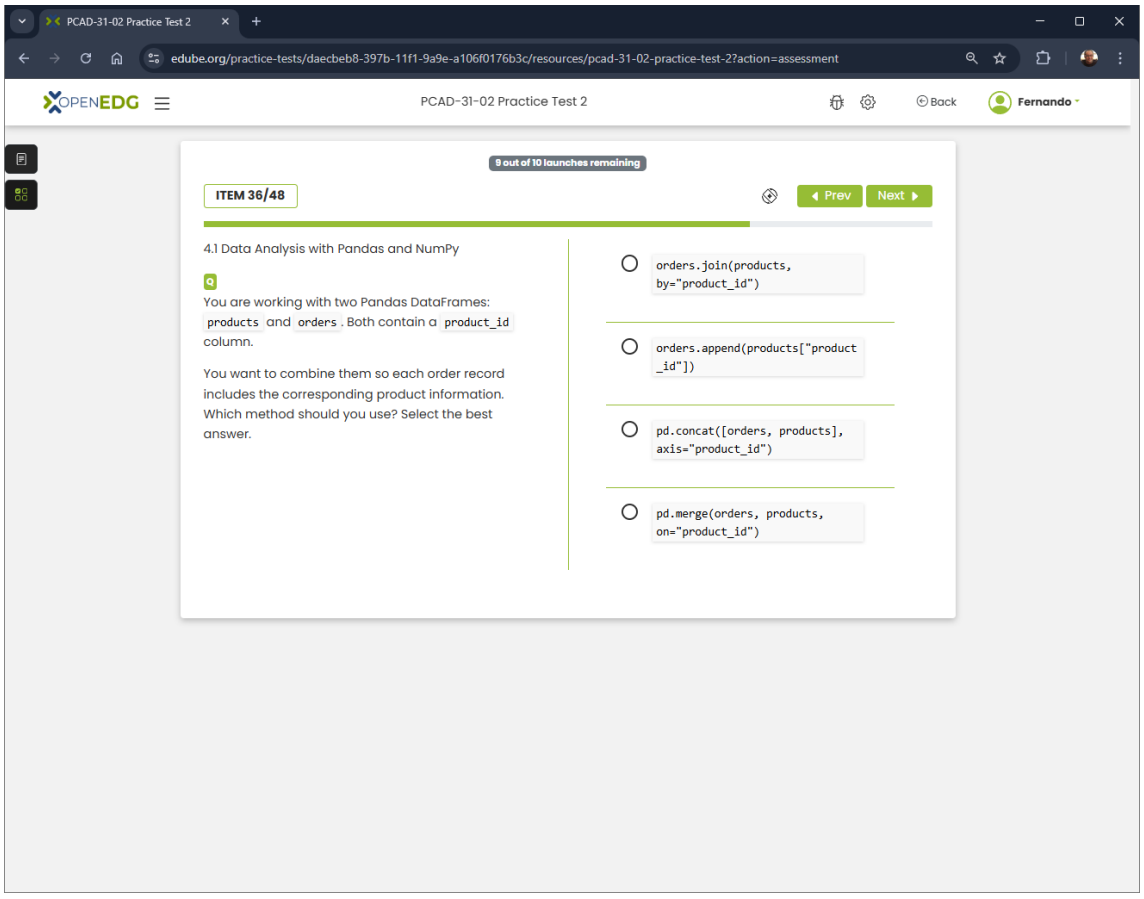
Which conclusions are supported by the heatmap? Select two best answers.

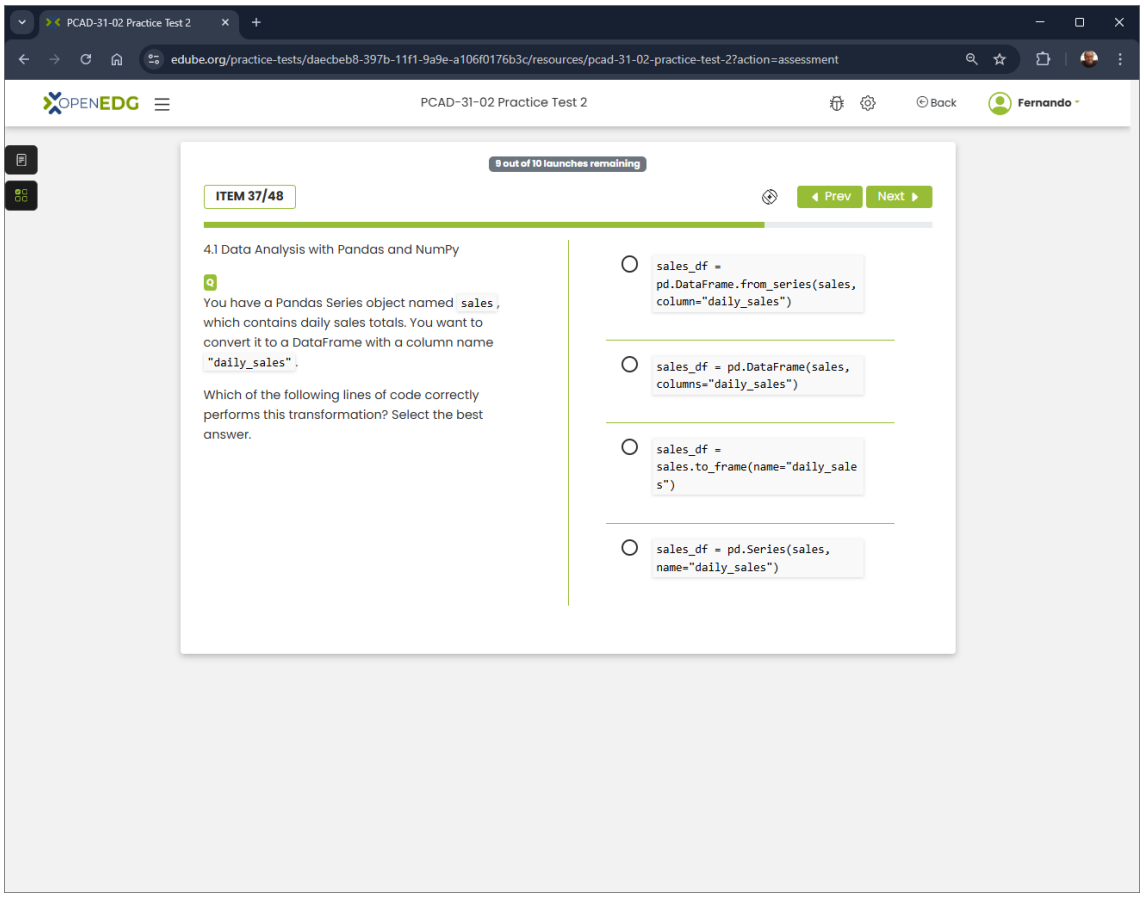
- [ ] Variable B shows a weak or no linear correlation with Variable D.
- [ ] High correlations in the heatmap confirm the dataset is normally distributed.
- [ ] There is a strong positive correlation between Variable A and Variable C.
- [ ] The heatmap proves that changes in Variable A cause changes in Variable B.
- [ ] All variable pairs have strong negative correlations, as shown by the dark blue cells.

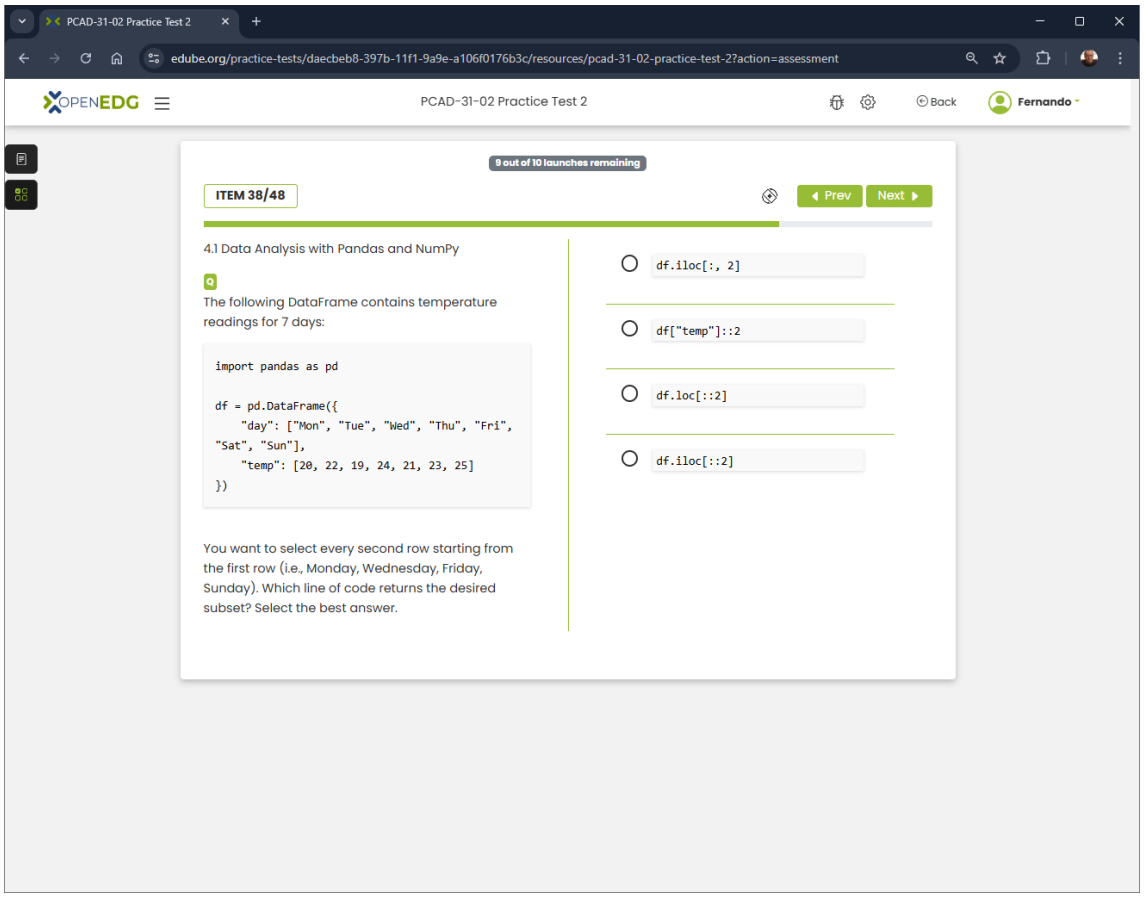

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 33/48
Prev
Next
3.2 Inferential Statistics
The following Python code performs a basic bootstrapping procedure on a dataset:
import numpy as np
data = [12, 15, 18, 20, 21]
means = []
for _ in range(1000):
sample = np.random.choice(data,
size=len(data), replace=True)
means.append(np.mean(sample))
result = np.mean(means)
What is the most likely purpose of this code? Select the best answer.
Calculate the population mean by summing all bootstrapped values.
Generate synthetic data for model training by duplicating original values.
Repeat the sampling to generate the true mean of the sample set.
Estimate the mean of the population by simulating sampling distributions from the original data.

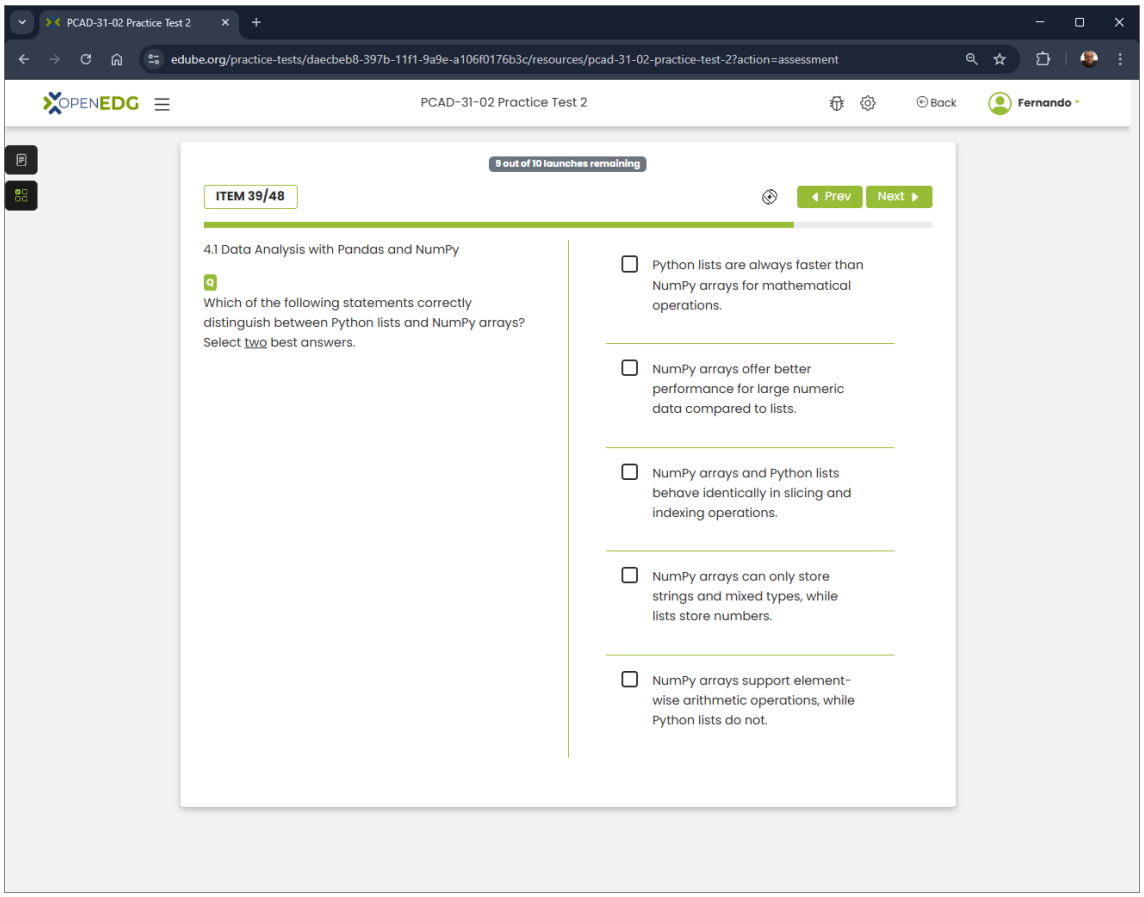

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 34/48
Prev
Next
3.2 Inferential Statistics
A data analyst fits a logistic regression model in Python using scikit-learn. The target variable is "is_fraud" (1 = fraudulent, 0 = not fraudulent). After fitting the model, they call model.predict_proba(X_test).
What does this function return? Select the best answer.
The log-odds of fraud for each observation.
The predicted probabilities for each class (0 and 1) for all test observations.
A binary decision matrix comparing predictions with actual test values.
The predicted binary classification labels for all test observations.

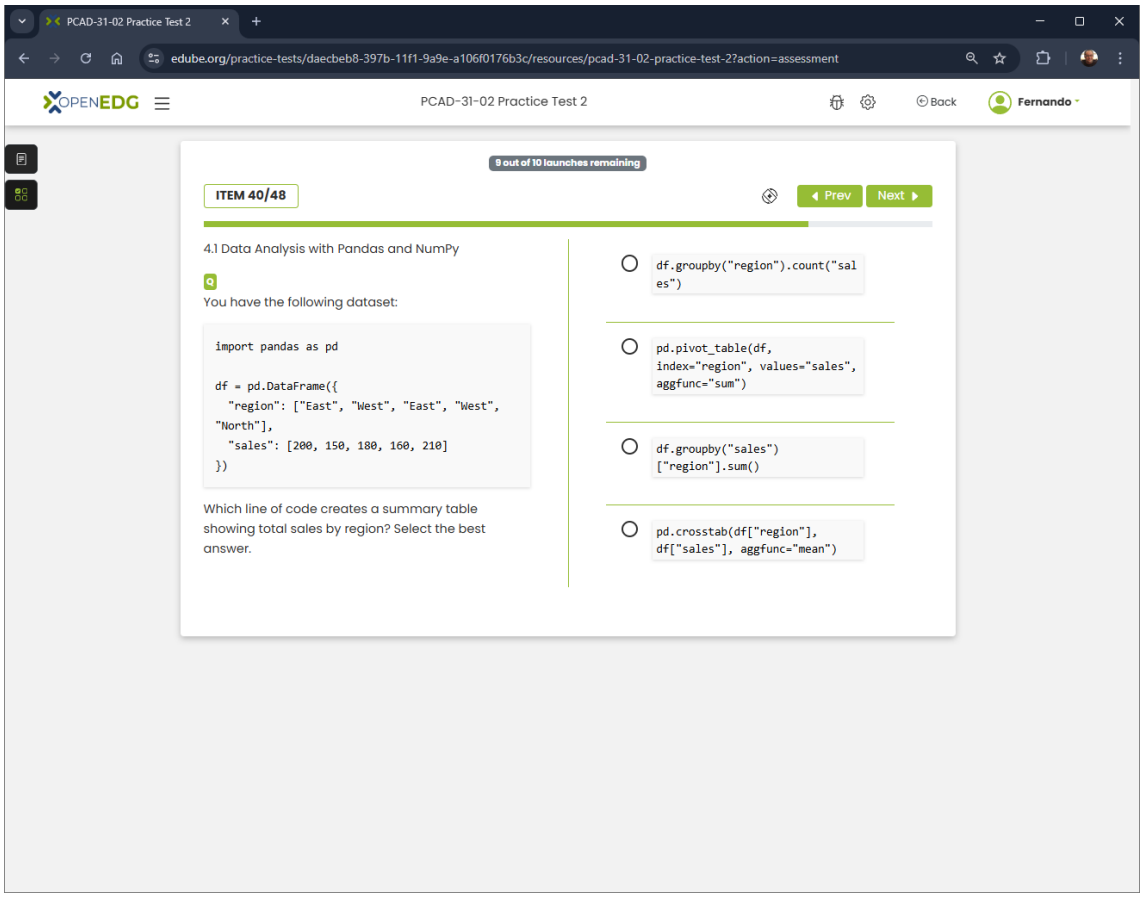

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 35/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
You are working with the following DataFrame:
import pandas as pd
df = pd.DataFrame({
"name": ["Alice", "Bob", None, "David"],
"age": [25, 30, 22, None]
})
Which line of code correctly removes rows with any missing values? Select the best answer.
df.dropna()
df.dropna(axis=1)
df.replace("NaN", "")
df.fillna(0)

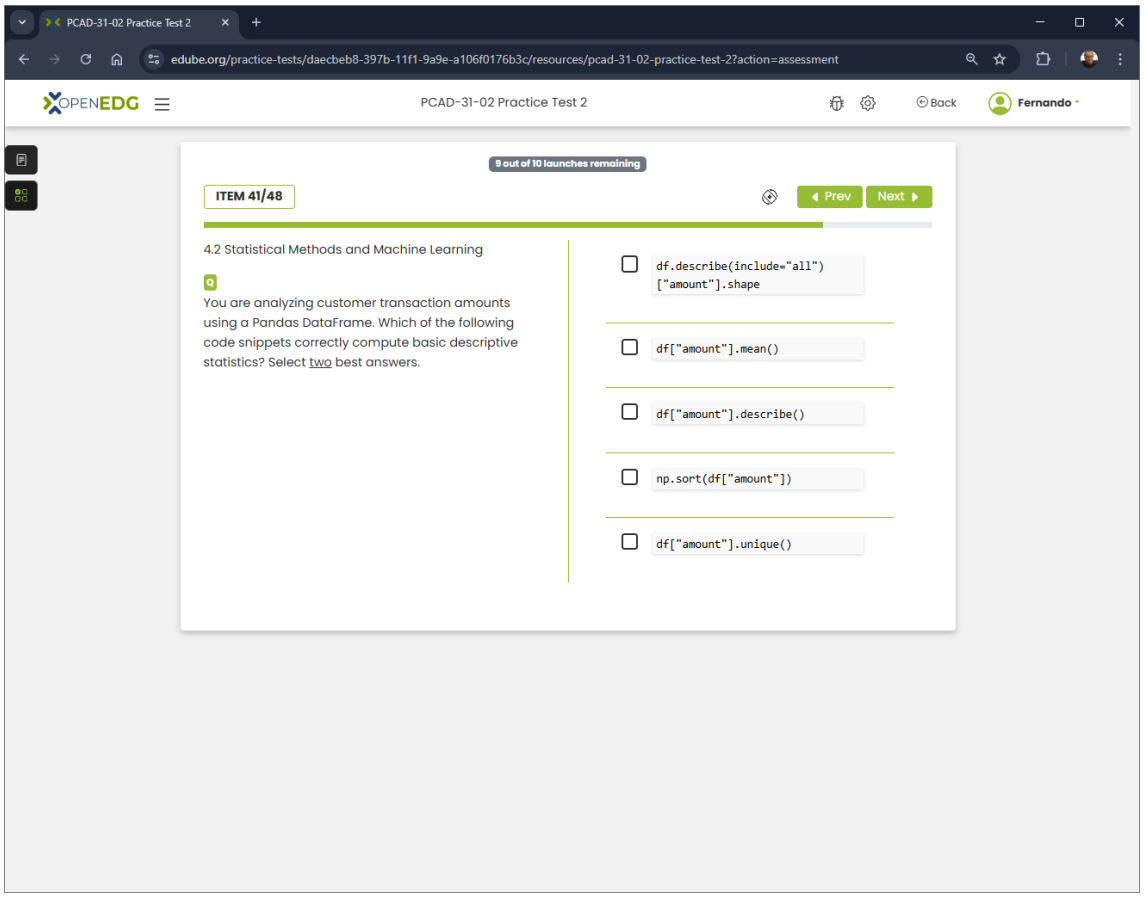

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 36/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
You are working with two Pandas DataFrames: products and orders. Both contain a product_id column.
You want to combine them so each order record includes the corresponding product information. Which method should you use? Select the best answer.
orders.join(products, by="product_id")
orders.append(products["product_id"])
pd.concat([orders, products], axis="product_id")
pd.merge(orders, products, on="product_id")

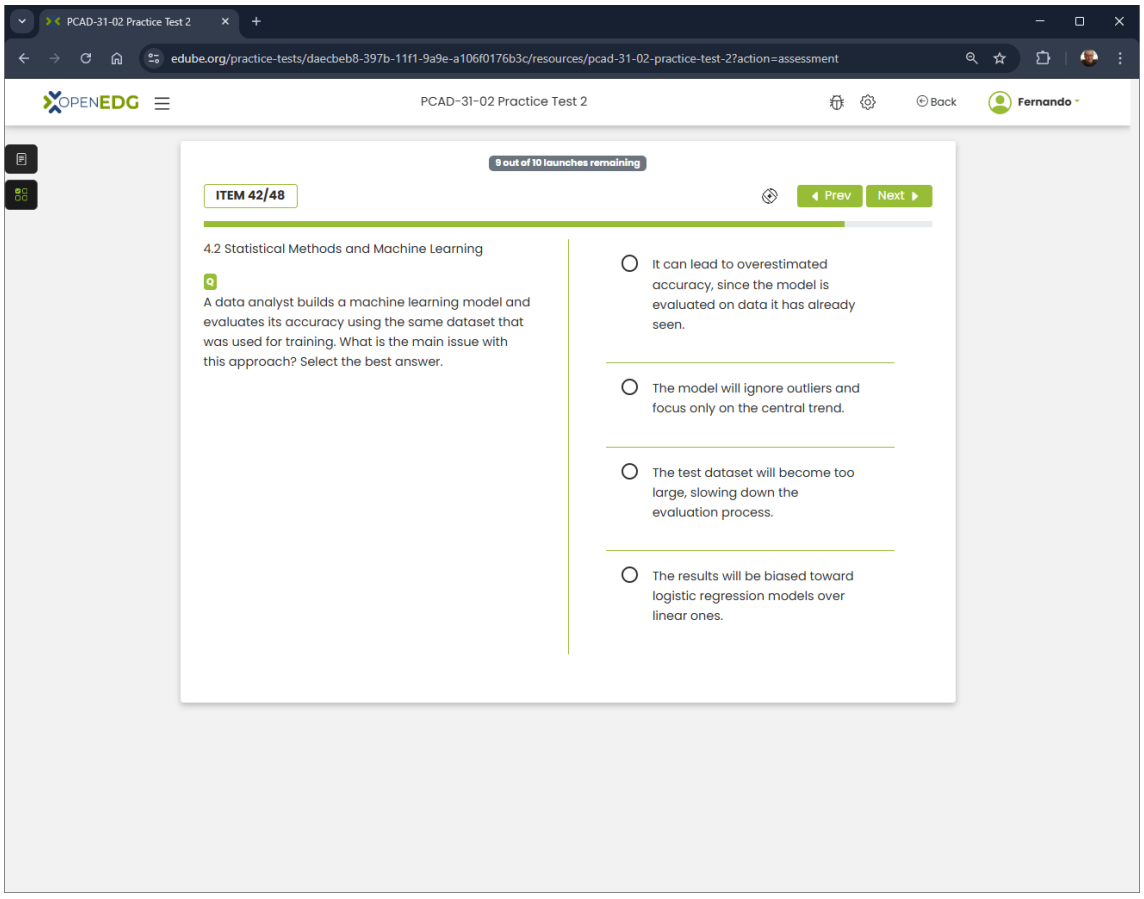

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 37/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
Q
You have a Pandas Series object named sales, which contains daily sales totals. You want to convert it to a DataFrame with a column name "daily_sales".
Which of the following lines of code correctly performs this transformation? Select the best answer.
sales_df = pd.DataFrame.from_series(sales, column="daily_sales")
sales_df = pd.DataFrame(sales, columns="daily_sales")
sales_df = sales.to_frame(name="daily_sales")
sales_df = pd.Series(sales, name="daily_sales")

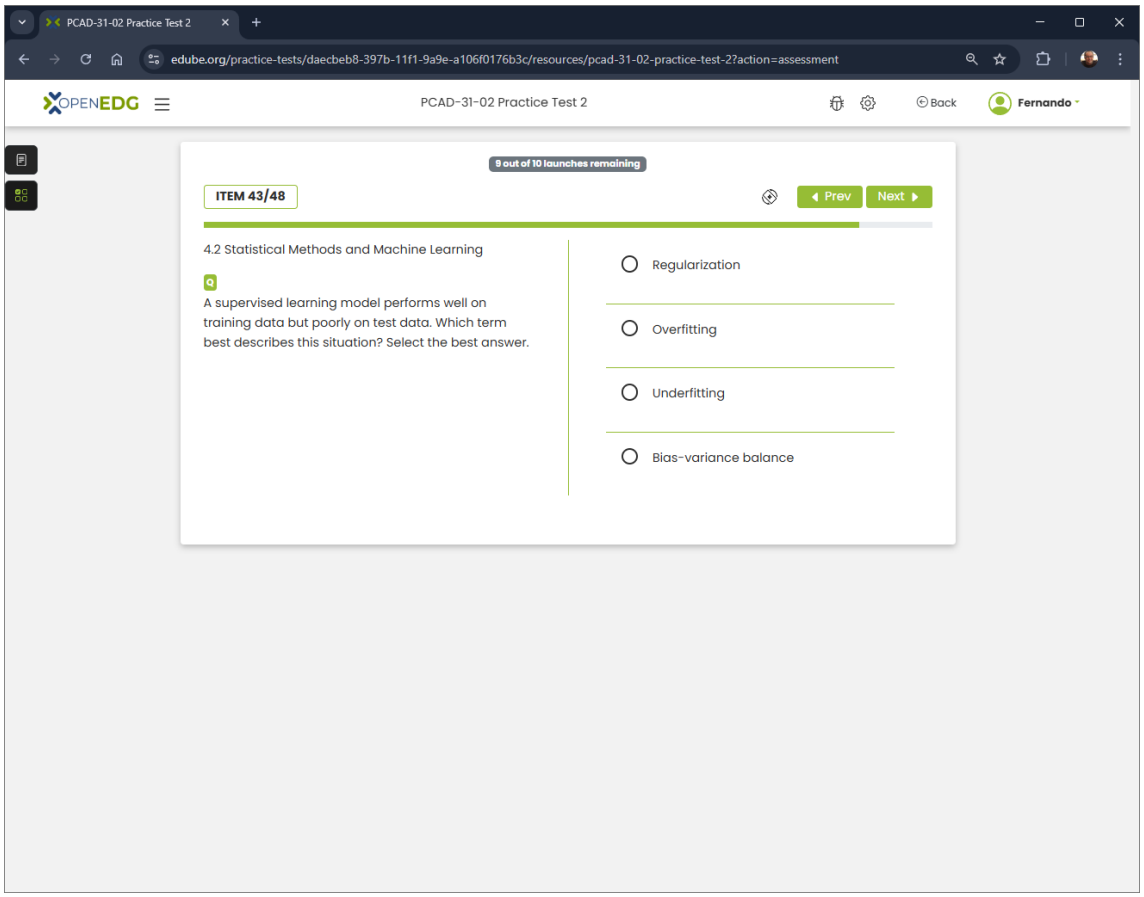

PCAD-31-02 Practice Test 2
edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment
OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 38/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
Q
The following DataFrame contains temperature readings for 7 days:
import pandas as pd
df = pd.DataFrame({
"day": ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"],
"temp": [20, 22, 19, 24, 21, 23, 25]
})
You want to select every second row starting from the first row (i.e., Monday, Wednesday, Friday, Sunday). Which line of code returns the desired subset? Select the best answer.
df.iloc[:, 2]
df["temp"]::2
df.loc[::2]
df.iloc[::2]

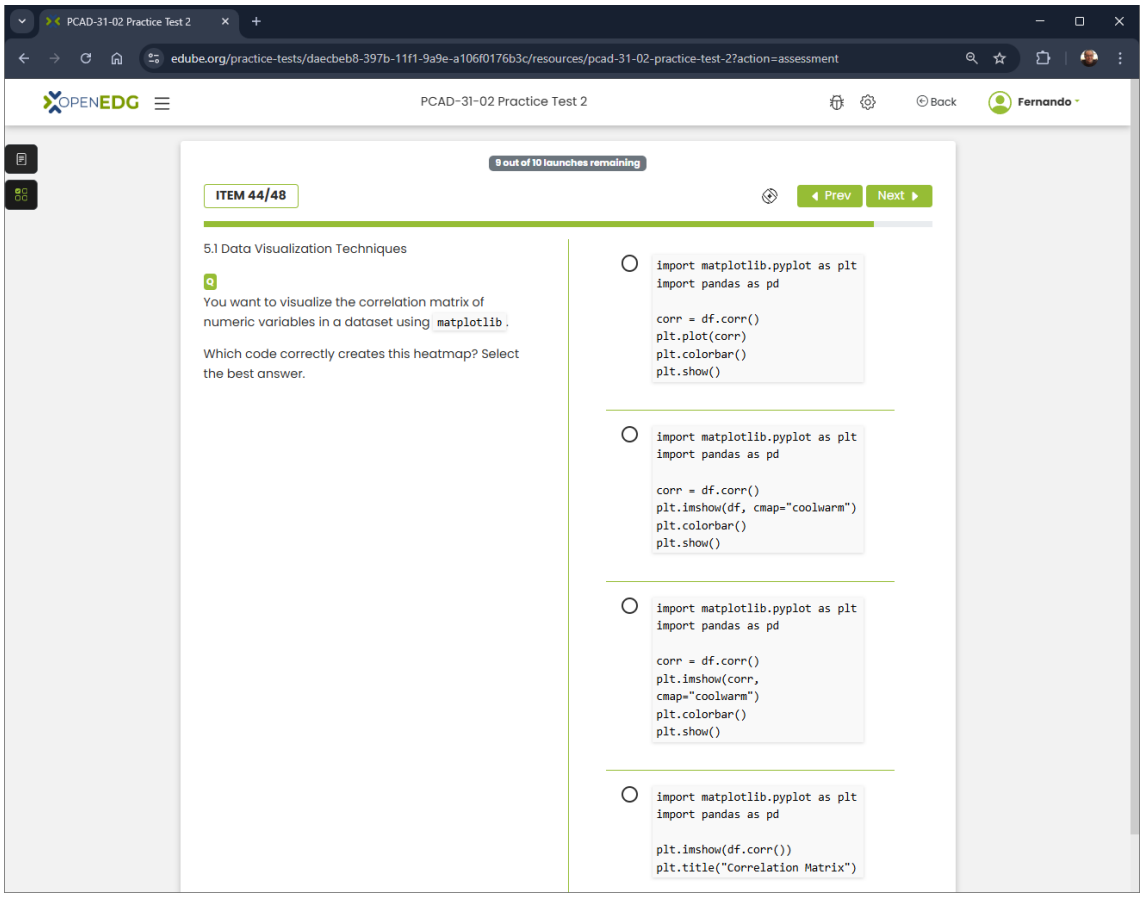

PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 39/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
Q
Which of the following statements correctly distinguish between Python lists and NumPy arrays? Select two best answers.
Python lists are always faster than NumPy arrays for mathematical operations.
NumPy arrays offer better performance for large numeric data compared to lists.
NumPy arrays and Python lists behave identically in slicing and indexing operations.
NumPy arrays can only store strings and mixed types, while lists store numbers.
NumPy arrays support element-wise arithmetic operations, while Python lists do not.


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 40/48
Prev
Next
4.1 Data Analysis with Pandas and NumPy
Q
You have the following dataset:
import pandas as pd
df = pd.DataFrame({
"region": ["East", "West", "East", "West", "North"],
"sales": [200, 150, 180, 160, 210]
})
Which line of code creates a summary table showing total sales by region? Select the best answer.
df.groupby("region").count("sales")
pd.pivot_table(df, index="region", values="sales", aggfunc="sum")
df.groupby("sales")["region"].sum()
pd.crosstab(df["region"], df["sales"], aggfunc="mean")


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 41/48
Prev
Next
4.2 Statistical Methods and Machine Learning
You are analyzing customer transaction amounts using a Pandas DataFrame. Which of the following code snippets correctly compute basic descriptive statistics? Select two best answers.
df.describe(include="all")["amount"].shape
df["amount"].mean()
df["amount"].describe()
np.sort(df["amount"])
df["amount"].unique()


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 42/48
Prev
Next
4.2 Statistical Methods and Machine Learning
A data analyst builds a machine learning model and evaluates its accuracy using the same dataset that was used for training. What is the main issue with this approach? Select the best answer.
It can lead to overestimated accuracy, since the model is evaluated on data it has already seen.
The model will ignore outliers and focus only on the central trend.
The test dataset will become too large, slowing down the evaluation process.
The results will be biased toward logistic regression models over linear ones.


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 43/48
Prev
Next
4.2 Statistical Methods and Machine Learning
A supervised learning model performs well on training data but poorly on test data. Which term best describes this situation? Select the best answer.
Regularization
Overfitting
Underfitting
Bias-variance balance


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 44/48
Prev
Next
5.1 Data Visualization Techniques
You want to visualize the correlation matrix of numeric variables in a dataset using matplotlib.
Which code correctly creates this heatmap? Select the best answer.
import matplotlib.pyplot as plt
import pandas as pd
corr = df.corr()
plt.plot(corr)
plt.colorbar()
plt.show()
import matplotlib.pyplot as plt
import pandas as pd
corr = df.corr()
plt.imshow(df, cmap="coolwarm")
plt.colorbar()
plt.show()
import matplotlib.pyplot as plt
import pandas as pd
corr = df.corr()
plt.imshow(corr,
cmap="coolwarm")
plt.colorbar()
plt.show()
import matplotlib.pyplot as plt
import pandas as pd
plt.imshow(df.corr())
plt.title("Correlation Matrix")

PCAD-31-02 Practice Test 2

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OPENEDG

PCAD-31-02 Practice Test 2

Back

Fernando

9 out of 10 launches remaining

ITEM 45/48

Prev Next

5.1 Data Visualization Techniques

Q

Two data analysis projects are being prepared for stakeholder presentations:

- **Project A** tracks monthly website traffic trends over a year and highlights peaks and dips in usage.
- **Project B** shows the proportional breakdown of product sales across four categories.

Which chart would be the most appropriate for each project? Select two best answers.

- [ ] Use the **scatter plot** for Project A to compare multiple categories in a single time period.
- [ ] Use the **bar chart** for Project B to analyze seasonal fluctuation in product prices.
- [ ] Use the **line chart** for Project A to emphasize changes and trends over time.
- [ ] Use the **pie chart** for Project B to highlight proportional differences in product sales.
- [ ] Use the **pie chart** for Project A to illustrate how traffic varies across weeks.

PCAD-31-02 Practice Test 2

edube.org/practice-tests/daecbeb8-397b-11f1-9a9e-a106f0176b3c/resources/pcad-31-02-practice-test-2?action=assessment

OPENEDG

PCAD-31-02 Practice Test 2

Back

Fernando

9 out of 10 launches remaining

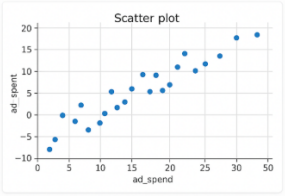
ITEM 46/48

Prev Next

5.1 Data Visualization Techniques

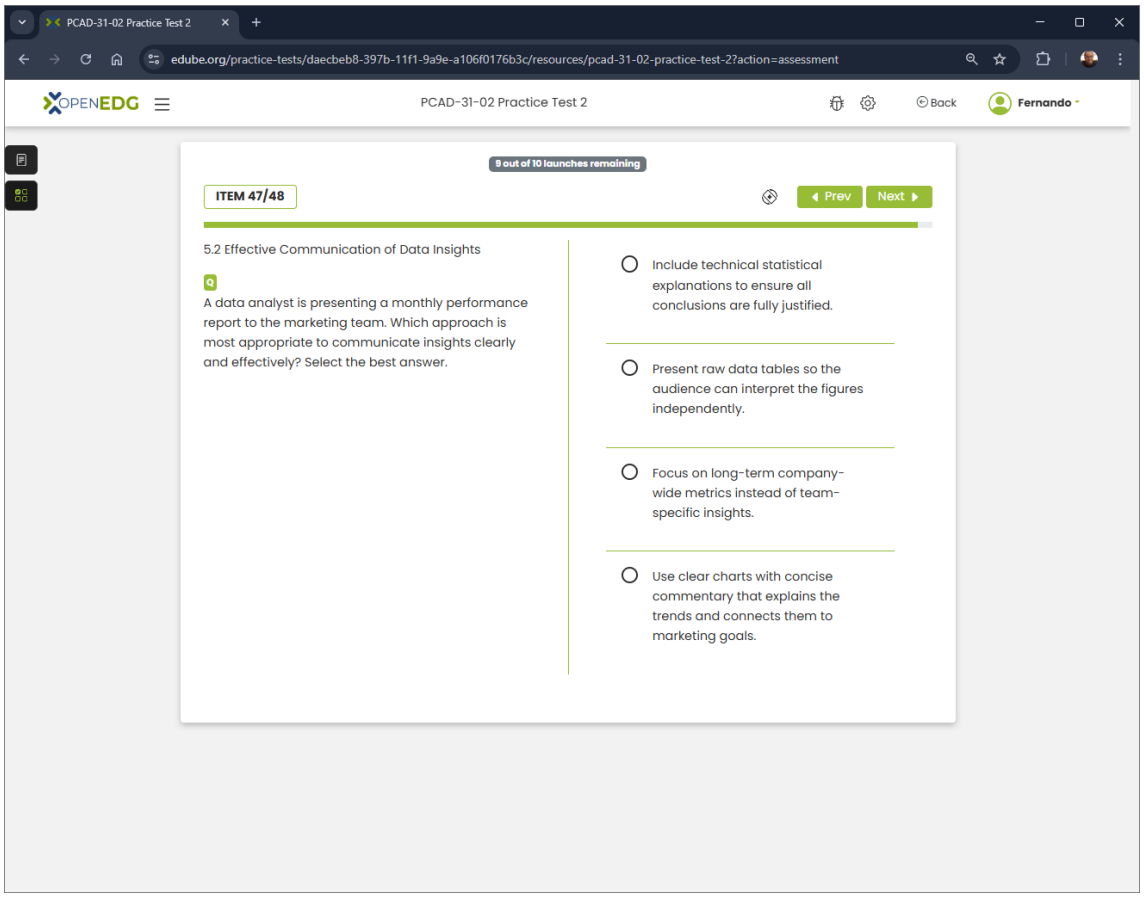
Q

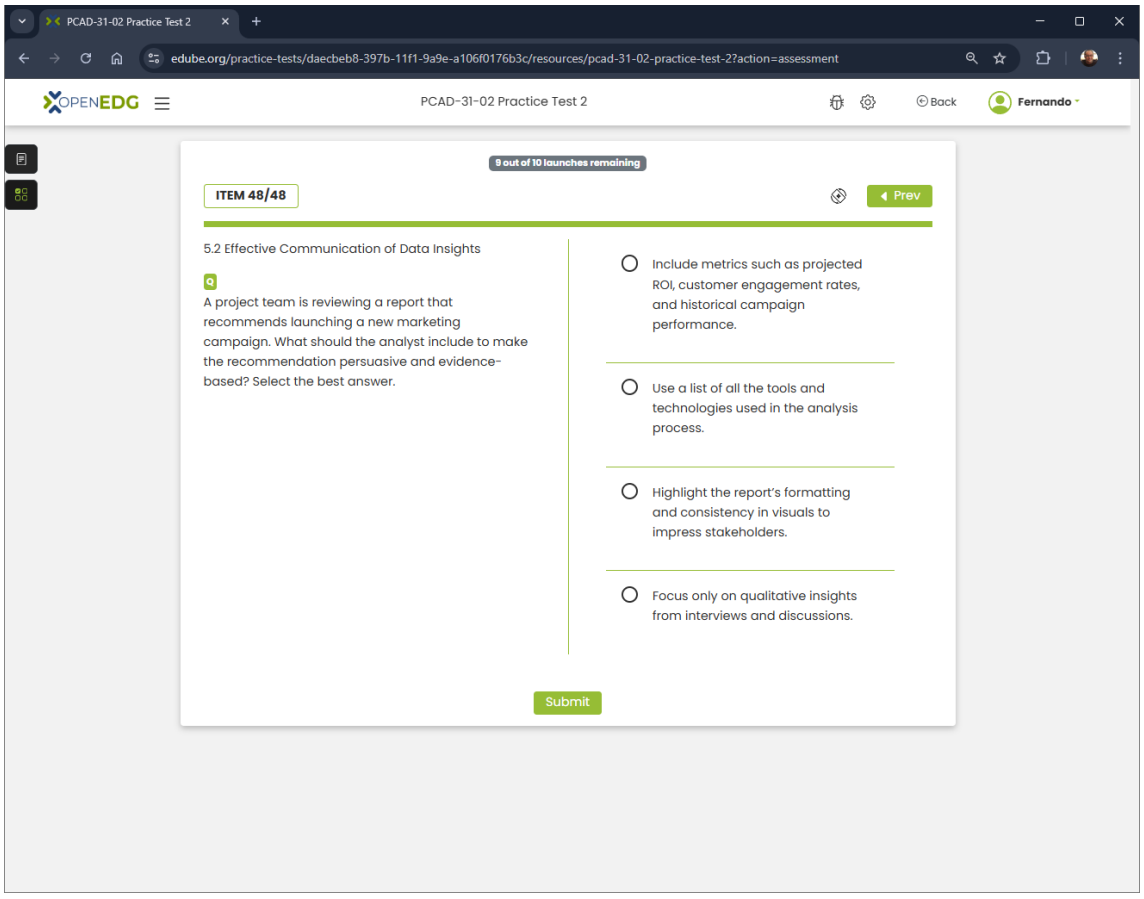
The scatterplot below displays the relationship between ad spend and sales:



Which annotations and refinements would improve this visualization? Select two best answers.

- [ ] Color the points green to better distinguish them from the grid
- [ ] Include a subtitle, e.g. "Correlation Between Ad Budget and Sales"
- [ ] Add axis titles, e.g. "Ad Spend ($)" and "Sales Revenue ($)"
- [ ] Incorporate a legend for clarity, e.g. "Sales Data"
- [ ] Remove the gridlines entirely to make the plot look cleaner


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 47/48
Prev
Next
5.2 Effective Communication of Data Insights
A data analyst is presenting a monthly performance report to the marketing team. Which approach is most appropriate to communicate insights clearly and effectively? Select the best answer.
Include technical statistical explanations to ensure all conclusions are fully justified.
Present raw data tables so the audience can interpret the figures independently.
Focus on long-term company-wide metrics instead of team-specific insights.
Use clear charts with concise commentary that explains the trends and connects them to marketing goals.


PCAD-31-02 Practice Test 2
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OPENEDG
PCAD-31-02 Practice Test 2
Back
Fernando
9 out of 10 launches remaining
ITEM 48/48
Prev
5.2 Effective Communication of Data Insights
A project team is reviewing a report that recommends launching a new marketing campaign. What should the analyst include to make the recommendation persuasive and evidence-based? Select the best answer.
Include metrics such as projected ROI, customer engagement rates, and historical campaign performance.
Use a list of all the tools and technologies used in the analysis process.
Highlight the report's formatting and consistency in visuals to impress stakeholders.
Focus only on qualitative insights from interviews and discussions.
Submit